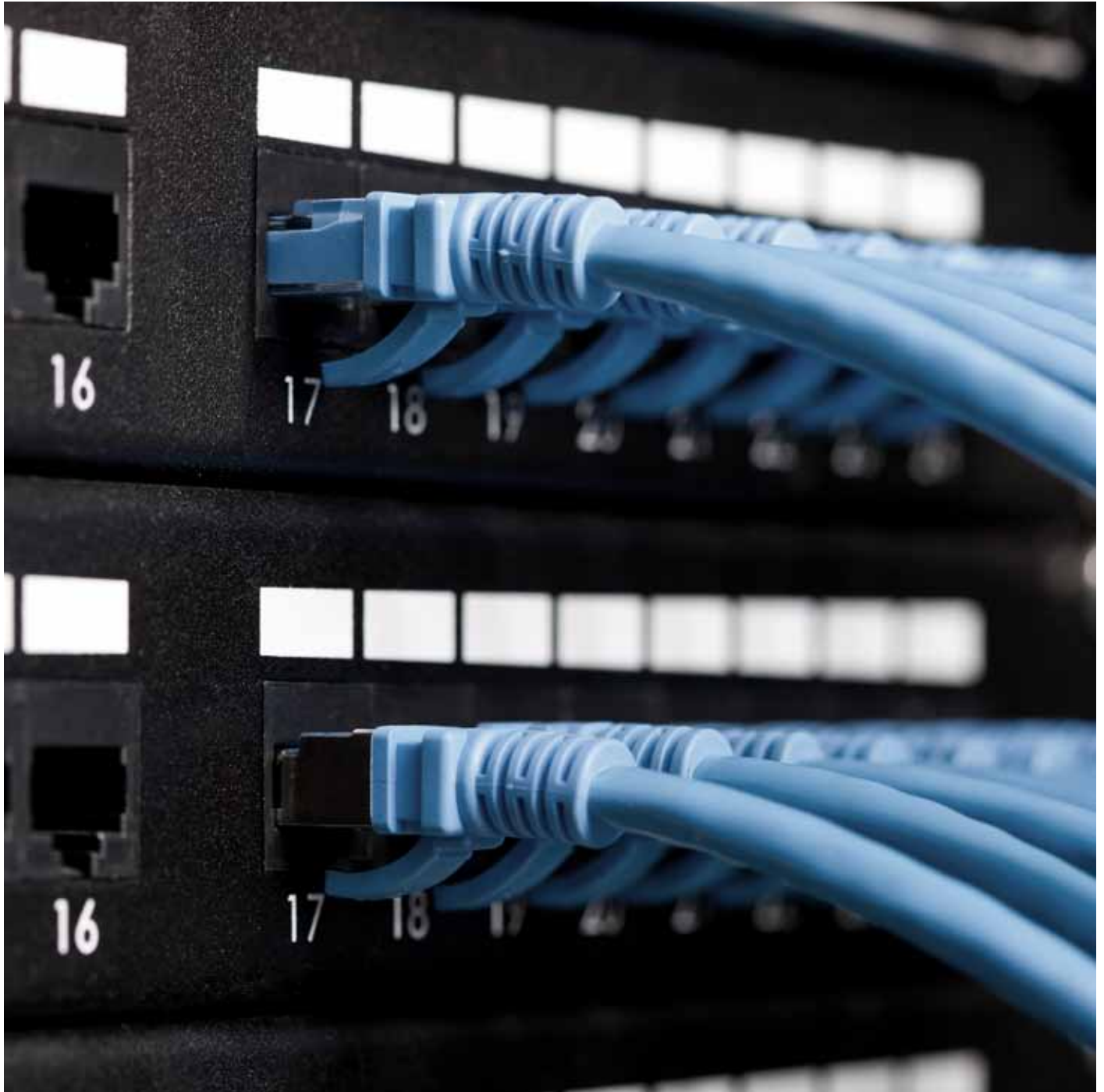




SimpleTM

지 명 원



LS Cable & System

THE WORLD BEST CABLE SOLUTION LEADER

LS Cable & System supplies various cables and materials used for power grids and communication networks around the world across all industries providing its top class technology and excellent quality. The company has also developed state of the art products, such as superconductors, HVDC and submarine cables that will lead the future energy industry.

LS spun off from LG in 2003 as a group specializing in electronics, electrical systems, energy and materials.



LS Cable & System

Transmission Cable
Distribution Cable
Submarine Cable
Telecommunication Cable
Industrial Cable
Industrial Material

LS ELECTRIC

Electric &
Automatic Equipments

LS-Nikko Copper

Copper Refinement

LS Mitron

Mechanical &
Electronic Parts

yesco
LNG

Ei
LPG

GAON
Power &
Communication
Cables



Keep your business *Simple*™

LS Simple™은 LS전선의 통합 배선 브랜드로 일반 주거 환경부터 데이터 센터를 비롯한 대규모 사업장까지 폭 넓은 분야에 통신 네트워크 솔루션을 제공하고 있습니다. 케이블은 물론 커넥터, 패치 패널, 관리 소프트웨어 등을 비롯한 Total Solution을 제공하여 고객의 Simple한 시스템 구성과 운영을 지원합니다.

LS Cable & System은 통신 인프라 구성에 있어 신뢰성, 보안성, 안전성을 갖춘 최적화된 Total Solution을 제공합니다. 또한, 고객에게 세계 최고 수준의 성능을 제공함으로써, 고객의 사업을 성장시키고 경쟁에서 우위를 가지도록 합니다.

50년 넘는 기간 동안 LS Cable & System은 기술의 진보를 이끌어 왔으며, 고객에게 현재는 물론, 미래의 요구에도 부응하는 Solution을 제공하여 왔습니다. LS Cable & System은 Cable부터 Connectors, Cable Assemblies, Patch Panels, Outlets, Infrastructure Management Solution을 제공하며, Backbone에서부터 Work Station까지 네트워크 전반에 걸쳐 최적의 Solution개발에 전념하고 있습니다.

Contents

Cables

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회사현황

회사명	엘에스전선(주)
대표이사	명노현
본사	경기도 안양시 동안구 호계동 1026-6 LSE타워 12~17층
홈페이지	www.lscns.com, www.lssimple.com
회사설립	62.05.15
사업분야	전력,통신,기기 및 소재 분야의 Total Solution Provider
세부 Solution	Power Transmission & Distribution (전력 송배전)
	Wind Power (풍력)
	Railway & Rolling Stock (철도 & 철도차량)
	Marine & Offshore (해양 & 선박)
	Automotive (자동차)
	Airport (공항)
	U-City (유시티)
	통합배선 시스템 (인빌딩)
	IBS (지능형 빌딩)
공장	구미공장 - 경북 구미시 공단동 190번지
	인동공장 - 경북 구미시 진평동 643번지
	동해공장 - 강원도 동해시 송정동
매출액	약 4,6조원
임직원	약 3,500명

회사연혁

1962.05 한국케이블공업 주식회사 설립, 구인회 사장 취임
1966.04 한국케이블공업 안양공장 준공
1966.09 주식회사 금성사로 합병
1969.10 금성전선 설립, 독립회사로 출범
1971.02 일본 히타치전선과 합작투자 및 포괄적 기술제휴계약 체결
1977.06 기업공개
1977.07 안양공장 수해 및 복구
1978.05 구미공장 준공
1979.10 구미공장, 국내 최초의 SCR시스템 준공
1982.08 안양 기술연구소 준공

1962 ~ 1982

한국의 대동맥으로 일어서다.

1983.10 한국중공업 군포공장 인수
1983.11 구미공장에 초고압 VCV타워 준공
1984.02 美 AT&T사와 광섬유 생산 합작회사 금성광통신 설립
1988.09 88서울올림픽 CATV시스템 구축 및 운영
1989.08 1000만 달러 규모 파키스탄 초고압 프로젝트 수주(최초의 턴키 수출)
1994.08 말레이시아 합작법인 LGM, LGE 공장 준공
1995.03 베트남에 광케이블 생산법인 VINA-GSC 설립
1996.08 세계 최대 규모의(303km, 9000만 달러) 이스라엘 초고압 프로젝트 수주
1997.06 베트남 합작법인 LG-VINA Cable 공장 준공
1997.05 구미시에 인동공장 준공
1999.12 노사화합 대상 대통령상 수상
2000.08 LG산전의 PCB용 초박막 도체 사업 인수

1983 ~ 2002

성장을 위한 변화를 추구하다.

2003.11 LG그룹으로부터 그룹분리, LG전선그룹 출범
2004.04 IN4S 프로젝트 킥오프, ERP 시스템 도입
2005.09 LS그룹 C선포식, LS전선으로 공식 출범
2005.03 진로산업(現 JS전선) 인수
2005.03 중국 우시 LS산업단지 준공
2007.12 베트남 호치민에 베트남 제2 전선공장 LSCV 준공
2008.05 경기도 안양시에 LS타워 준공
2008.07 ㈜LS 지주회사 체제 출범, LS엠펙트 분할
2008.08 북미 최대 전선회사 수페리어에식스(Superior Essex) 인수
2009.02 진도~제주간 122km 해저전력망 연계 사업 수주
2009.08 중국 호북용딩홍치전기(現 LS홍치전선) 인수
2009.11 중동해시에 해저케이블 공장 준공
2011.06 경기도 이천변전소 초전도실증사업 준공
2011.11 후베이성(湖北省) 우한(武漢)시에 R&D센터 설립
2012.04 인도 하리아나(Haryana)주에 LSCI 전력공장 준공
2012.04 경기도 군포시에 R&D센터 준공
2012.10 미국 노스캐롤라이나주 타버러(Tarboro)시에 전력케이블 공장 준공
2012.12 중국 LS홍치전선 VCV타워 준공
2013.02 국내 최초 유럽 해저케이블 시장 진출
2013.03 국내 최초 남미 해저케이블 시장 진출
2013.04 국내 최초 초고압직류송전(HVDC) 케이블 해외(덴마크) 수출
2013.12 LS아이앤디(주) 기업 분할
2015.01 신비전 'LS전선 Way' 선포
2015.01 세계 최초 직류 초전도 케이블 실증 시험 완료
2015.05 LS전선아시아(베트남 법인 지주회사) 설립
2016.09 LS전선아시아(베트남 법인 지주회사) 유가증권시장 상장
2017.03 미국 전력케이블 생산법인 설립
2018.05 세계 최초 HVDC 케이블 공인 인증
2019.05 폴란드 생산법인 준공
2020.04 동해 해저케이블 2공장 준공

2002 ~ 현재

백년기업을 향한 꿈을 펼치다.

◎ 등록증

사업자 등록증	08
사업자 등록증명	09
공장 등록증 (구미)	10
공장 등록증 (인동)	11

사업자 등록증

사업자 등록증명

국세청
nts.go.kr

사업자 등록증

(법인사업자)

등록번호 : 138-81-47634

법인명(단체명) : 엘에스전선(주)

대표자 : 명노현

개업연월일 : 2008년 07월 01일 법인등록번호 : 134111-0168357

사업장소재지 : 경기도 안양시 동안구 엘에스로 127(호계동, LS타워 12~17층)

본점소재지 : 경기도 안양시 동안구 엘에스로 127(호계동, LS타워 12~17층)

사업의종류 : ☒ 제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

제조업 ☒ 제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

제조,도소매,건설 ☐ 제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

제조,도소매,건설 ☐ 제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

제조,도소매 ☐ 제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

제조,도소매 ☐ 제조업 ☐ 판매업 ☐ 서비스업 ☐ 기타업

발급사유 : 정정 (별지 출력)

사업자 단위 과세 적용사업자 여부 : 여(✓) 부() (적용일자: 2008년 06월 23일)

전자세금계산서 전용 전자우편주소 :

2020년 02월 10일

동 안 양 세 무 서 장

국세청
nts.go.kr

(1 / 1)

사업자 등록증명

Certificate of Business Registration
(법인사업자)
(Corporate Taxpayer)

발급번호 Issuance number	6164-926-3160-068	처리기간 Processing period	즉시 Immediately
상호(법인명) Name of company	엘에스전선(주) LS Cable & system Ltd.		
사업자등록번호 Business registration number	138-81-47634		
성명(대표자) Name of representative	명노현 Myung Roe Hyun		
주민(법인)등록번호 Resident(Corporation) registration number	134111-0168357		
사업장소재지 Business Address	경기도 안양시 동안구 엘에스로 127(호계동, LS타워 12~17층) 12-17 Floor, LS Tower, 127, LS-ro, Dongan-gu, Anyang-si, Gyeonggi-do		
개업일 Date of business commencement	2008년(Year) 07월(Month) 01일(Day)		
사업자등록일 Date of business registration	2008년(Year) 06월(Month) 23일(Day)		
업태 Business type	제조업/제조업 Manufacturing/Manufacturing		
종목 Business item	제조업/제조업 Manufacture of Other Insulated Wire and Cable 산업용고무제품 Manufacture of Synthetic Rubber		
공동사업자 Joint business owner	성명(법인명) Name(Name of company)	주민(사업자)등록번호 Resident(Business) registration no.	
	해당사항 없습니다 (No Data)		

위와 같이 증명합니다.
I certify that above information is true and correct to the best of my knowledge and belief.
출위내용은 발급일 현재 상황으로서 추후 변경될 수 있습니다.
The information is true as of the issue date of this certificate and it can be subject to change.

접수번호 Receipt No.	500658146478	2017년 3월 27일 Year Month Day
담당부서 Department	민원봉사실 Taxpayer Service Center	
담당자 Staff in Charge	임승수 Im Seungsu	
연락처 Telephone No.	031-389-8404	

동 안 양 세 무 서 장
Head of Donganyang District Tax Office

국세청

* 본 증명은 국세청 홈택스(www.hometax.go.kr)에서 「민원증명 원본확인」 메뉴를 통해 문서발급번호로 위·변조 여부를 확인하거나, 문서 하단과 바코드로 확인해 주십시오. 다만 문서발급번호를 통한 확인은 발급일로부터 90일까지 가능합니다.
* 본 증명은 홈택스(www.hometax.go.kr)대인 온라인 서비스를 통해 발급된 증명서입니다.

공장 등록증 (인동)

문서확인번호: 1491-2045-8637-4145 (신청인 : 엘에스전선)

■ 산업집적활성화 및 공장설립에 관한 법률 시행규칙 [별지 제8호의2서식] <개정 2012.10.5> 공장성립관리지원시스템(www.femis.go.kr)에서도 신청할 수 있습니다.

공장등록증명(신청서)

※ 바탕색이 어두운 난은 신청인이 적지 않으며, []에는 해당하는 곳에 √표를 합니다. (앞쪽)

접수번호	접수일	처리기간	즉시
신청인	회사명 LS전선(주)2공장	전화번호 054) 469-7710	
	대표자 성명 명노현, 윤재인	생년월일(법인등록번호) 134111-0168357	
	대표자주소(법인소재지) 경기도 안양시 동안구 엘에스로 127 (호계동, LS타워)		

등록 내용	공장소재지 도로명 : 경상북도 구미시 3공단2로 238 (전통동, 엘에스전선주식회사인동공장) 지번 : 경상북도 구미시 전량동 643번지	지목 공장용지	보유구분 자가 [√] 임대 []
	공장등록일 2000-04-18	사업시작일 1969-10-25	종업원수 남:340 여:44
	공장의 업종(분류번호) 광섬유 케이블 제조업 외 2 층 (28301,24222,28302)		
	공장부지면적 247,327.400	제조시설면적 82,175.470 m ²	부대시설면적 11,050.590 m ²

등록 조건	등록변경 · 증설 등 기재사항 변경내용(변경 날짜 및 내용)	공장관리번호 : 110111000396503
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2016-06-02

「산업집적활성화 및 공장설립에 관한 법률 시행규칙」 제12조의3에 따라 위와 같이 공장등록증명서를 신청합니다.

2017년 4월 3일

신청인

엘에스전선 (서명 또는 인)

구미시장

귀하

구비서류	없음	수수료	원
처리절차			
신청서작성 신청인	→ 접수 지리기관	→ 등록 여부 확인 지리기관	→ 결재 지리기관
→ 공장등록 증명서 발급 지리기관		→ 정보 지리기관	

「산업집적활성화 및 공장설립에 관한 법률」 제16조([] 제1항 · [] 제2항 · [] 제3항)에 따라 위와 같이 등록된 공장임을 증명합니다.

2017년 4월 3일

구미시장

210mmX297mm(일반용지 70g/㎡(제활용종))

김소연 / 4월3일 16:29

◆본 증명서는 인터넷으로 발급되었으며, 민원24(minwon.go.kr)의 인터넷발급문서진위확인 메뉴를 통해 위·변조 여부를 확인해 주십시오.(발급일로부터 90일까지) 또한 문서하단의 바코드도 진위확인(스캐너용 문서확인프로그램 또는 민원24 앱)을 하실 수 있습니다.

◎ 시스템 인증서

ISO 9001 (구미,인동)	14
ISO 14001 (구미,인동)	16
ISO 16949 (구미,인동)	18
OHSAS 18001 (구미,인동)	20
TL 9000 (인동)	22



CERTIFICATE





This is to certify that

LS Cable & System Ltd.
Gumi Plant

228, Suchul-daero, Gumi-si
Gyeongsangbuk-do, 39369
Republic of Korea

with the organizational units/sites as listed in the annex

has implemented and maintains a **Quality Management System**.

Scope:
The design and manufacture of copper telecommunication cables, power cables, power cable accessories, overhead AL conductors, OPGW (optical fiber composite ground wires) and busduct systems. Manufacturing of bare conductors, copper rods and OFC (Oxygen Free Copper).

Through an audit, documented in a report, it was verified that the management system fulfills the requirements of the following standard:

ISO 9001 : 2015

Certificate registration no.	20004748 QM15
Date of original certification	1993-07-16
Valid from	2018-12-17
Valid until	2021-12-16
Date of certification	2018-12-03




DQS Inc.



Brad McGuire
Managing Director



Accredited Body: DQS Inc., 1130 West Lake Cook Road, Suite 340, Buffalo Grove, IL 60089 USA
 Administrative Office: DQS Korea LLC., #302, ACE Techno Tower 10-cha,
 196, Gasan digital 1-ro, Geumcheon-gu, Seoul, 08502, Republic of Korea

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Annex to certificate
Registration No. 20004748 QM15

LS Cable & System Ltd.
Gumi Plant

228, Suchul-daero, Gumi-si
Gyeongsangbuk-do, 39369
Republic of Korea



Location	Scope
20005028 LS Cable & System Ltd. 12-17th Fl. LS Tower 127 LS-ro, Dongan-gu Anyang-si Gyeonggi-do, 14119 Republic of Korea	Customer-related process.
20001787 LS Cable & System Ltd. Indong Plant 238, 3gongdan 2-ro Gumi-si Gyeongsangbuk-do, 39395 Republic of Korea	The design and manufacture of appliance wiring materials. Design, development and management of outsourced hardware for telecommunication cabling and micro coaxial cable assembly manufactures.

DQS Inc.



Brad McGuire
Managing Director



Accredited Body: DQS Inc., 1130 West Lake Cook Road, Suite 340, Buffalo Grove, IL 60089 USA
 Administrative Office: DQS Korea LLC., #302, ACE Techno Tower 10-cha,
 196, Gasan digital 1-ro, Geumcheon-gu, Seoul, 08502, Republic of Korea

2 / 2



CERTIFICATE





This is to certify that

LS Cable & System Ltd.
Gumi Plant
228, Suchul-daero
Gumi-si, Gyeongsangbuk-do, 39369
Republic of Korea

with the organizational units/sites as listed in the annex

has implemented and maintains an **Environmental Management System**.

Scope:
The environmental activities and supporting processes, associated with the Design and manufacture of copper telecommunication cables, optical fibers and cables, power cable, power cable accessories, busduct system, distributed feedback laser diodes, aluminium extruded products, extruded tubes, electronic wires and cables, low and high voltage electric cables and the manufacture of magnet wires, aluminum billet and rods, aluminum tubes, copper rods and automotive wires

Through an audit, documented in a report, it was verified that the management system fulfills the requirements of the following standard:

ISO 14001 : 2015

Certificate registration no. 20004748 UM15

Date of certification 2018-09-13

Valid until 2021-09-12




DQS Inc.



Brad McGuire
Managing Director



Accredited Body: DQS Inc., 1130 West Lake Cook Road, Suite 340, Buffalo Grove, IL 60089 USA
Administrative Office: DQS Korea LLC., #302, ACE Techno Tower 10-cha,
196, Gasan digital 1-ro, Geumcheon-gu, Seoul, 08502, Republic of Korea

1 / 2



Annex to certificate
Registration No. 20004748 UM15

LS Cable & System Ltd.
Gumi Plant
228, Suchul-daero
Gumi-si, Gyeongsangbuk-do, 39369
Republic of Korea



Location	Scope
20001787 LS Cable & System Ltd. Indong Plant 238, 3Gongdan2-ro, Gumi-si Gyeongsangbuk-do, 39395 Republic of Korea	The environmental activities and supporting processes, associated with the Design and manufacture of copper telecommunication cables, optical fibers and cables, power cable, power cable accessories, busduct system, distributed feedback laser diodes, aluminium extruded products, extruded tubes, electronic wires and cables, low and high voltage electric cables and the manufacture of magnet wires, aluminium billet and rods, aluminium tubes, copper rods and automotive wires
20004719 LS Cable & System Ltd. Donghae Plant 215, Daedong-ro, Donghae-si Gangwon-do, 25812 Republic of Korea	The environmental activities and supporting processes, associated with The design and manufacture of submarine cables and system, and industrial specialty cables and system.



This annex (edition: 2018-09-13) is only valid in connection with the above-mentioned certificate.

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CERTIFICATE



This is to certify that

LS Cable & System Ltd.
 Gumi Plant
 228, Suchul-daero
 Gumi-si, Gyeongsangbuk-do, 39369
 Republic of Korea

has implemented and maintains a **Quality Management System**.

Scope:
 Manufacturing of magnet wires.

An audit, conducted and documented in a report, has verified that this quality management system fulfills the requirements of the following International Automotive Standard:

IATF 16949:2016
 (without product design)

Certificate registration no.	20004748 IATF16	 2-IAQ-QMC-01001
Main certificate registration no.	20004748 IATF16	
Issuing date	2018-09-06	
This certificate is valid until	2021-09-05	
IATF No.	0330349	

For and on behalf of DQS


 Nurani S. Venkatachalam
 Corporate Program Manager Automotive, DQS Holding GmbH


 Michael Drechsel
 Managing Director, DQS Holding GmbH

IATF Contract Office: DQS Holding GmbH, Konrad-Adenauer-Allee 8-10, 61118 Bad Vilbel, Germany

1 / 2 2017-03-09



CERTIFICATE



This is to certify that

LS Cable & System Ltd.
 Indong Plant
 238, 3Gongdan2-ro
 Gumi-si, Gyeongsangbuk-do, 39395
 Republic of Korea

has implemented and maintains a **Quality Management System**.

Scope:
 Manufacturing of aluminum micro multiport and round tubes, aluminum-alloyed billet, bars and rods, wires, cables, extruded tubings and sleeveings.

An audit, conducted and documented in a report, has verified that this quality management system fulfills the requirements of the following International Automotive Standard:

IATF 16949:2016
 (without product design)

Certificate registration no.	20001787 IATF16	 2-IAQ-QMC-01001
Main certificate registration no.	20004748 IATF16	
Issuing date	2018-07-22	
This certificate is valid until	2021-07-21	
IATF No.	0317846	

For and on behalf of DQS


 Nurani S. Venkatachalam
 Corporate Program Manager Automotive, DQS Holding GmbH


 Michael Drechsel
 Managing Director, DQS Holding GmbH

IATF Contract Office: DQS Holding GmbH, Konrad-Adenauer-Allee 8-10, 61118 Bad Vilbel, Germany

1 / 2 2017-03-09



CERTIFICATE





This is to certify that

LS Cable & System Ltd.
Gumi Plant
228, Suchul-daero
Gumi-si, Gyeongsangbuk-do, 39369
Republic of Korea

with the organizational units/sites as listed in the annex

has implemented and maintains an
Occupational Health and Safety Management System.

Scope:
The occupational health and safety activities and supporting processes associated with the design and manufacture of copper telecommunication cables, optical fibers and cables, power cable, power cable accessories, busduct system, distributed feed back laser diodes, aluminum extruded products, extruded tubes, electronic wires and cables, low and high voltage electric cables, and the manufacture of magnet wires, aluminum billet and rods, aluminum tubes, copper rods and automotive wires.

Through an audit, documented in a report, it was verified that the management system fulfills the requirements of the following standard:

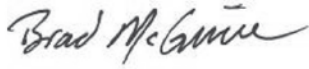
BS OHSAS 18001 : 2007



Certificate registration no.	20004748 BSOH
Date of original certification	2004-09-07
Date of certification	2019-09-06
Valid until	2021-03-11



DQS Inc.



Brad McGuire
Managing Director





Accredited Body: DQS Inc., 1500 McConnor Parkway, Suite 400, Schaumburg, IL 60173 USA
Administrative Office: DQS Korea LLC., #302, ACE Techno Tower 10-cha,
196, Gasan digital 1-ro, Geumcheon-gu, Seoul, 08502, Republic of Korea

1 / 2



Annex to certificate
Registration No. 20004748 BSOH

LS Cable & System Ltd.
Gumi Plant
228, Suchul-daero
Gumi-si, Gyeongsangbuk-do, 39369
Republic of Korea





Location

20001787
LS Cable & System Ltd.
Indong Plant

238, 3Gongdan2-ro, Gumi-si
Gyeongsangbuk-do, 39395
Republic of Korea



20004719
LS Cable & System Ltd.
Donghae Plant

215, Daedong-ro, Donghae-si
Gangwon-do, 25812
Republic of Korea



Scope

The occupational health and safety activities and supporting processes associated with the design and manufacture of copper telecommunication cables, optical fibers and cables, power cable, power cable accessories, busduct system, distributed feedback laser diodes, aluminium extruded products, extruded tubes, electronic wires and cables, low and high voltage electric cables and the manufacture of magnet wires, aluminium billet and rods, aluminium tubes, copper rods and automotive wires.





This annex (edition: 2019-09-06) is only valid in connection with the above-mentioned certificate.

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CERTIFICATE





This is to certify that

LS Cable & System Ltd.
Indong Plant

238, 3Gongdan2-ro, Gumi-si
Gyeongsangbuk-do, 39395
Republic of Korea

with the organizational units/sites as listed in the annex

has implemented and maintains a **Quality Management System**.

Scope:
The design and manufacture of optical fibers and optical fiber cables.

Product Categories:
3.1.1.2.1 Fiber Optic Cable
8.5.1 Optical Fiber

Through an audit documented in a report, in accordance with QuEST Forum Code of Practice for TL 9000 Registrars and ISO/IEC 17021 Conformity Assessment - Requirements for bodies providing audit and certification of management systems, it was verified that the management system fulfills the requirements of the following standards:

TL 9000-H R6.0/R5.5 ISO 9001 : 2015

TL Id number	TL1868
Certificate registration no.	20001787 TLR6
Date of original certification	2002-01-23
Date of certification	2018-09-14
Valid until	2021-09-13



DQS Inc.



Brad McGuire
Managing Director

Accredited Body: DQS Inc., 1130 West Lake Cook Road, Suite 340, Buffalo Grove, IL 60089 USA
Administrative Office: DQS Korea LLC., #302, ACE Techno Tower 10-cha,
196, Gasan digital 1-ro, Geumcheon-gu, Seoul, 08502, Republic of Korea



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Annex to certificate
Registration No. 20001787 TLR6

LS Cable & System Ltd.
Indong Plant

238, 3Gongdan2-ro, Gumi-si
Gyeongsangbuk-do, 39395
Republic of Korea





Location

20005029
LS Cable & System Ltd.

LS Tower, 127, LS-ro, Dongan-gu
Anyang-si, Gyeonggi-do, 14119
Republic of Korea

Scope

Sales and marketing



This annex (edition: 2018-09-14) is only valid in connection with the above-mentioned certificate.

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www.lssimple.co.kr | 25

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인증번호: 제 17-0139 호



제 품 인 증 서

1. 제조업체명 : 엘에스케이블 System Vietnam Co., Ltd.
2. 대표자성명 : 송우성
3. 공장소재지 : NHON TRACH 2-LOC KHANG INDUSTRIAL ZONE HIEP PHUOC VILLAGE, NHON TRACH DISTRICT, DONG NAI PROVINCE, VIETNAM
4. 인증제품
 - 가. 표준명 : 근거리 통신 케이블
 - 나. 표준번호 : KS C 3342
 - 다. 종류·등급·호칭 또는 모델 :
 - CM, 6등급 4P HDPE F/UTP
 - CM, 6등급 4P HDPE SF/UTP
 - CM, 6등급 4P HDPE U/UTP
 - CM, 6A등급 4P HDPE U/UTP
 - CM, 5E등급, 25P, HDPE, U/UTP
 - CM, 5E등급, 4P, HDPE, U/UTP
 - CMR, 5E등급, 4P, HDPE, U/UTP. 끝.

「산업표준화법」 제17조 제1항에 따른 인증심사를 실시한 결과 한국 산업표준(KS)과 인증심사기준에 적합하므로, 「산업표준화법」 제15조 및 같은 법 시행규칙 제10조 제1항에 따라 위와 같이 한국산업표준(KS)에 적합함을 인증합니다.

2020 년 03 월 18 일



한국표준협회



1. 최초 인증일 : 2017-05-24
2. 차기심사 완료기한 : 2020-05-23
3. 최종 변경일 : 2020-03-18 종류 추가



ONLINE CERTIFICATIONS DIRECTORY

DVBI.E352897

Local Area Network Cable Verified for Transmission Performance in Accordance with National or International Specifications

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Local Area Network Cable Verified for Transmission Performance in Accordance with National or International Specifications

See General Information for Local Area Network Cable Verified for Transmission Performance in Accordance with National or International Specifications

LS CABLE & SYSTEM LTD

GUMI PLANT
190 KONGDAN-DONG
GUMI-SHI, GYEONGSANGBUK-DO 730-708 REPUBLIC OF KOREA

Data transmission cable

CM - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3, 5E, 6, 6A.
- NEMA WC 66 Category 6.
- UL Data Transmission Program (CAT 5).

CM Patch Cable - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 5E, 6.

CM-LS - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 6A.

CMG - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3, 5E, 6, 6A.
- NEMA WC 66 Category 6.
- UL Data Transmission Program (CAT 5).

CMP - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3, 5E, 6, 6A.
- NEMA WC 66 Category 6.
- UL Data Transmission Program (CAT 5).

CMR - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3, 5E, 6, 6A.
- NEMA WC 66 Category 6.
- UL Data Transmission Program (CAT 5).

CMR Patch Cable - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 5E, 6.

CMX - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3, 5E, 6, 6A.
- NEMA WC 66 Category 6.
- UL Data Transmission Program (CAT 5).

CMX Outdoor - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3.
- UL Data Transmission Program (CAT 5).

CMX Outdoor-CM - Listed and verified in accordance with:

- ANSI/TIA-568-C.2 Category 3.
- UL Data Transmission Program (CAT 5).

CMX Outdoor-CMG - Listed and verified in accordance with:

E352897

http://database.ul.com/cgi-bin/XYV/template/LISEXT/1FRAME/showpage.html?name=DVBI.E352897&ccshorttitle=Local+Area+Network+Cable+Verified... 1/2

UL 인증 (통합배선 자재)

2017. 4. 3. DUXR.E245354 - Communications-, Audio/Video-, Data- and Other Signaling-circuit Accessories

Keystone jack cover inserts, Model(s) LS-FP10-SB-BB-XX (XX=jack type, M/J, SC, LC, MTRJ, D2 or D1)

Keystone jacks, Model(s) LG-3013A-VH8-C5eABL-XX, LG-3013A-VH8-C6ABL-XX, LS-MJ-EU-UC5E-WH, LS-MJ-EU-UC6-WH, LS-MJ-SC5E-WH-RIDC, LS-MJ-SC5E-XX, LS-MJ-SC6-ERI, LS-MJ-SC6A-ERI, LS-MJ-SC6-WH-RIDC, LS-MJ-SC6-XX, LS-MJ-SC6A-WH-TL, LS-MJ-UC5E-RI-BK, LS-MJ-UC5E-XX

Keystone jacks, Model(s) LS-MJ-UC5E-XX-ERI, LS-MJ-UC5E-XX-ERI-EX, LS-MJ-UC6-XX-ERI, LS-MJ-UC6-XX-ERI-EX, LS-MJ-UC6A-XX-ERI, LS-MJ-UC6A-XX-ERI-EX, PT-MJ-UC5E-XX-RI, PT-MJ-UC6-XX-RI, PT-MJ-UC6A-XX-RI

Keystone jacks, Model(s) LS-MJ-UC5E-xx-TL (^), LS-MJ-UC5E-XX-WP1, LS-MJ-UC5E-XX-WP10-y-zz (a), LS-MJ-UC5E-XX-WP11-y-zz a), LS-MJ-UC5E-XX-WP12-y-zz (a), LS-MJ-UC5E-XX-WP2, LS-MJ-UC5E-XX-WP4-XX, LS-MJ-UC6-BL-S, LS-MJ-UC6-IV-WP1(4103744), LS-MJ-UC6-WH-S, LS-MJ-UC6-XX, LS-MJ-UC6-xx-TL (^), LS-MJ-UC6-XX-WP1, LS-MJ-UC6-XX-WP13-y-zz (a), LS-MJ-UC6-XX-WP2, LS-MJ-UC6-XX-WP5-XX, LS-MJ-UC6A-XX, LS-MJ-UC6A-XX-SI, LS-MJ-UC6E-BL-PLUS, LS-MJ-UC6E-WH-PLUS, LS-MJ-UC6E-WH-PLUS-EX, LS-MJ-XX, LS-MJ10-UC5E-XX-TL-y (a), LS-MJ10-UC6-XX-TLS-y-zz (a)

Keystone jacks, Model(s) PT-MJ-CU5E-BL, PT-MJ-CU5E-WH, PT-MJ-CU5E-YL, PT-MJ-CU5E-BK, PT-MJ-CU5E-RD, PT-MJ-CU5E-GN, PT-MJ-CU5E-IV, PT-MJ-CU5E-GY, PT-MJ-CU6-BL, PT-MJ-CU6-WH, PT-MJ-CU6-YL, PT-MJ-CU6-BK, PT-MJ-CU6-RD, PT-MJ-CU6-GN, PT-MJ-CU6-IV, PT-MJ-CU6-GY

Modular IDC patch panels, Model(s) LG-BPXX-A86N-C6MABL, LG-BPXX-N86A-C5eABL

Modular Patch Cord, Model(s) LS-PC-SC5E-XX-YYY (n), LS-PC-SC5E-XX-YYY-AW (n), LS-PC-SC6-XX-YYY (n), LS-PC-SC6-XX-YYY-AW (n), LS-PC-SC6A-XX-YYY (n), LS-PC-UC5E-XX-YYY (n), LS-PC-UC5E-XX-YYY-AW (n), LS-PC-UC6-XX-YYY (n), LS-PC-UC6-XX-YYY-AW (n), LS-PC-UC6A-XX-YYY (n)

Modular patch panels, "HC-BPXX Series", Model(s) PT-PP-UC5E-12PM, PT-PP-UC5E-16PM, PT-PP-UC5E-24PM, PT-PP-UC5E-48PM, PT-TPP-25PM, PT-TPP-50PM

Modular patch panels, Model(s) LS-PP-SC5E-24P, LS-PP-SC6-24P, LS-PP-UC5E-12P, LS-PP-UC5E-16P, LS-PP-UC5E-24P, LS-PP-UC5E-48P, LS-PP-UC6-24P, LS-PP-UC6-48P, LS-PP-XX, LS-TPP-25P, LS-TPP-50P, LS-TPP-XX

Patch Cord, Model(s) HC-STP8XX, HC-UTP80 SERIES

Patch cords, Model(s) LS-110PC10-USC5E-02-XX-YYY (b)

Patch cords, Model(s) LS-110PC10-USC5E-04-XX-YYY (b)

Patch cords, Model(s) LS-110PC10-USC5E-08-XX-YYY (b)

Patch cords, Model(s) LS-110PCC-UC6-4-XX-YYY (XX=Color, YYY=Length)

Patch cords, Model(s) LS-IPCC-SC6(L)-XX-XXX, LS-IPCC-SC6A(L)-XX-XXX (XX=length)

Patch cords, Model(s) LS-IPCC-UC6(L)-XX-XXX, LS-IPCC-UC6A(L)-XX-XXX, LS-IPCI-UC6(A)(L), LS-IPCS-UC6(A)(L), LS-IPCA-UC6(L)-XX-XXX

Patch cords, Model(s) LS-PC-SC5E-XX-XXX, LS-PC-SC6-XX-XXX, LS-PC-SXX, LS-PC-UC5E-XX-XXX, LS-PC-UC6-XX-XXX, LS-PC-UC6A-XX-XXX, LS-PC-UXX, LS-PC10-UC6-XX-YYY (b)

Patch cords, Model(s) PT-PC-SXX, PT-PC-SC5EV-XX-XXX, PT-PC-SC6V-XX-XXX

Patch cords, Model(s) PT-PC-UXXV, PT-PC-UC5EV-XX-XXX, PT-PC-UC6V-XX-XXX

Patch panels, Model(s) LS-PP-24P-EX, LS-PP-UC6A-24P-WM-SI, LS-PP10-UC5E-XXP-A@, LS-PP10-UC5E-XXP-B@, LS-PP10-UC5E-XXP@, LS-PP10-UC6-XXP@, LS-PP11-UC5E-XXP@

Patch panels, Model(s) PT-PP-EP-24PP-WM, PT-PP-EP-48PP-WM

patch panels, Model(s) PT-PP-UC6-24PM, PT-PP-UC6-48PM

Shielded keystone jack modules, Model(s) LS-MJ10-SC5E-TL, LS-MJ10-SC6-TL

Snap-in inserts, Model(s) PT-AI-US-BI-1, PT-AI-US-BNC-1

Snap-in inserts, Model(s) PT-AI-US-LC-SM, PT-AI-US-LC-MM

Snap-in inserts, Model(s) PT-AI-US-SC-SM, PT-AI-US-SC-MM

Snap-in inserts, Model(s) PT-AI-US-ST-SM, PT-AI-US-ST-MM

Snap-in inserts, Model(s) PT-AI-US-TV-1

Surface mount boxes, Model(s) LG-SMB-1-VH8-C5eABL-XX, LG-SMB-1-VH8-C6ABL-XX, LG-SMB-2-VH8-C5eABL-XX, LG-SMB-2-VH8-C6ABL-XX, LS-SMB-1PORT, LS-SMB-2PORT, LS-SMB-4PORT, LS-SMB-6PORT, LS-SMB-XX

Surface mount boxes, Model(s) PT-SMB-1PORT, PT-SMB-2PORT, PT-SMB-4PORT, PT-SMB-6PORT

Surface plate assemblies, Model(s) LS-SMJ10-UC6-XPORT (X=port, 1, 2, 4 or 6)

Surface plate assemblies, Model(s) LS-SPA10-UC5E-X (X=port, 1, 2, 4 or 6)

Surfacemount jacks, Model(s) LS-SMJ10-UC5E-XPORT (X=port, 1, 2, 4 or 6)

Termination tools, Model(s) LS-CBLSPR-XX, LS-STOOL

W-1DC Wiring blocks, Model(s) 4-Pair10-U

Wall plate module inserts, Model(s) LS-WMB10-1, LS-WMB10-2

Wall plates, Model(s) LS-FP-US-1PORT, LS-FP-US-2PORT, LS-FP-US-4PORT, LS-FP-US-6PORT, LS-FP-US-XX

Wiring blocks, Model(s) LS-110WB-100P-W, LS-110WB-100P-W/O, LS-110WB-25P-W, LS-110WB-25P-W/O, LS-110WB-50P-W, LS-110WB-50P-W/O

Wiring blocks, Model(s) PT-110WB-UC5E-100P-W/O, PT-110WB-CU5E-100P-W

Wiring blocks, Model(s) PT-110WB-UC5E-200P-W/O, PT-110WB-CU5E-200P-W

UL 인증 (통합배선 자재)

2017. 4. 3. DUXR.E245354 - Communications-, Audio/Video-, Data- and Other Signaling-circuit Accessories

Wiring blocks, Model(s) PT-110WB-UC5E-25P-W/O, PT-110WB-UC5E-25P-W

Wiring blocks, Model(s) PT-110WB-UC5E-300P-W/O, PT-110WB-CU5E-300P-W

Wiring blocks, Model(s) PT-110WB-UC5E-50P-W/O, PT-110WB-UC5E-50P-W

(a) - Where XX denotes color, y denotes data or voice and zz denotes door color.

(b) - Where XX denotes color (WH, BL, YL, RD, GN, GY) and YYY denotes meter (005(0.5m),010(1m),etc.).

(n) - XX denotes Color:WH=White, BL=Blue, YL=Yellow, RD=Red, GN=Green, GY=Gray, BK=Black. YYY denotes Meter: 030(3ft),050(5ft),070(7ft),100(10ft),150(15ft),200(20ft),250(25ft)

(^)- (Remark) xx : Color Code ,

@ - Where XX denotes the number of ports (12, 24, 48...).

NOTE - The above units are followed by suffixes denoting the number of contacts or parts in the plugs or jacks, size of snap-in outlets or the number of parts in the patch panels whether shielded or nonshielded, keyed or nonkeyed. Intended for indoor use only on the equipment side of a Listed primary telephone protector.

XX - XX denotes colors, where BK represents black, BL represents blue, GN represents green, GR represents grey, OR represents orange, RD represents -red, WH represents white and YL represents yellow.)

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UL 인증 (모듈라 플러그)

2017. 4. 3. DUXR2.E245354 - Communications-, Audio/Video-, Data- and Other Signaling-circuit Accessories - Component

 **ONLINE CERTIFICATIONS DIRECTORY**

DUXR2.E245354
Communications-, Audio/Video-, Data- and Other Signaling-circuit Accessories - Component

[Page Bottom](#)

Communications-, Audio/Video-, Data- and Other Signaling-circuit Accessories - Component

[See General Information for Communications-, Audio/Video-, Data- and Other Signaling-circuit Accessories - Component](#)

LS CABLE & SYSTEM LTD E245354
12~17F LS TOWER
127 LS-RO DONGAN-GU
ANYANG-SI, GYEONGGI-DO 14119 REPUBLIC OF KOREA

Modular plugs, Model(s) LS-MP-C5E-RJ45, LS-MP-SC5E-RJ45-TRBOOT, LS-MP-SC6-RJ45-TRBOOT-A, LS-MP-SC6-RJ45-TRBOOT-B, LS-MP-SXX, LS-MP-UC5E-RJ45, LS-MP-UC6-RJ45, LS-MP-UC6-RJ45-K, LS-MP-UXX, PT-MP-SXX

Modular plugs, Model(s) PT-MP-UXX, PT-MP-UC5E-RJ45, PT-MP-UC6-RJ45, PT-MP-C5e-RJ45

   and
Marking: Company name or trademark
model designation on the product or on the smallest unit container in which the product is packaged.
[Last Updated](#) on 2016-11-23

[Questions?](#) [Print this page](#) [Terms of Use](#) [Page Top](#)

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UL 인증 (광케이블)

Certificate of Compliance

Certificate Number	20110214-E203488	Page 1 of 1
Report Reference	E203488, 1999 July 16	
Issue Date	2011 February 14	

 **Underwriters Laboratories**

Issued to: **LS CABLE LTD**
643 JINPYEONG-DONG
GUMI-SHI GYEONGBUK-DO 730-360 KOREA

This is to certify that representative samples of **OPTICAL FIBER CABLE**
OFNP, OFNR, OFN

Have been investigated by Underwriters Laboratories in accordance with the Standard(s) indicated on this Certificate.

Standard(s) for Safety: **UL 1651 - STANDARD FOR OPTICAL FIBER CABLE - Edition 3**

Additional Information: **See UL On-line Certification Directory at WWW.UL.COM for additional information.**

Only those products bearing the UL Listing Mark should be considered as being covered by UL's Listing and Follow-Up Service.

The UL Listing Mark generally includes the following elements: the symbol UL in a circle,  with the word "LISTED"; a control number (may be alphanumeric) assigned by UL; and the product category name (product identifier) as indicated in the appropriate UL Directory.

Look for the UL Listing Mark on the product

William R. Carney
Director, North American Certification Programs
Underwriters Laboratories Inc.
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For questions, please contact a local UL Customer Service Representative at <http://www.ul.com/global/eng/pages/corporate/contactus/>

CC-Link IE Field

Certificate

This is to certify that
the following product of
LS Cable&system Ltd. has passed
a CC-Link IE Field Network
Recommended Product Test

Description of product: CC-Link IE Field Cable (Flexible type)

Type: SFP-E-C5G-E1ZP-X0.5004P/GY(LS SIMPLE)

Date of Issue March 24, 2017

Reference Number 121



Chairman

 **CLPA** CC-Link Partner Association

CC-Link IE Field

Certificate

This is to certify that
the following product of
LS Cable&system Ltd. has passed
a CC-Link IE Field Network
Recommended Product Test

Description of product: CC-Link IE Field Cable(Fixed type)

Type: SFP-E-C5G-E1ZN-X 0.5X004P/GY

Date of Issue March 24, 2017

Reference Number 122



Chairman

 **CLPA** CC-Link Partner Association



Compliance Statement

Data communication cable, Category 5e

Company
LS Cable & System Vietnam Co., Ltd.
Nhon Trach 2-Loc Khang Industrial Zone
Nhon Trach District
Dong Nai Province
Vietnam

Product description
Unscreened cable (U/UTP) characterised up to 100 MHz
Horizontal floor wiring cable, 100 Ω
4 twisted pairs
Flame retardant, halogen free

Product identification
Cat. 5e UTP LSZH

Generic cabling and cabling components standards - Category 5e cable requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- IEC 61156-5:2012 (Ed. 2.1)
- EN 50173-1:2018 / EN 50173-2:2018
- EN 50288-3-1:2013
- TIA-568.2-D:2018

Standards for flammability, halogen acidity, and smoke performance

- IEC 60332-1-1:2015 (Ed. 1.1) / IEC 60332-1-2:2015 (Ed. 1.1)
- IEC 60754-2:2011 (Ed. 2.0)
- IEC 61034-1:2013 (Ed. 3.1) / IEC 61034-2:2013 (Ed. 3.1)

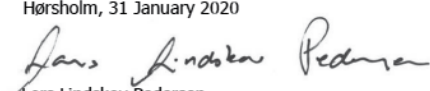
Technical report
T931777-01, DANAK-19/20915

EC Cabling product ID
5006

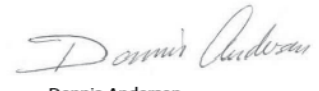
CS valid until
1 September 2020

This product has been tested by FORCE Technology and complies with the requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance schedule, which implies that FORCE Technology performs a quality audit of the manufacturer's production and QA sites. The maintenance testing of the product is performed on a sample basis.

Hørsholm, 31 January 2020



Lars Lindskov Pedersen
Test Manager



Dennis Andersen
Head of Department



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No. 2020-128



Compliance Statement

Data communication cable, Category 6

Company
LS Cable & System Vietnam Co., Ltd.
Nhon Trach 2-Loc Khang Industrial Zone
Nhon Trach District
Dong Nai Province
Vietnam

Product description
Unscreened cable (U/UTP) characterised up to 250 MHz
Horizontal floor wiring cable, 100 Ω
4 twisted pairs
Flame retardant, halogen free

Product identification
Cat. 6 UTP LSZH

Generic cabling and cabling components standards - Category 6 cable requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- IEC 61156-5:2012 (Ed. 2.1)
- EN 50173-1:2018 / EN 50173-2:2018
- EN 50288-6-1:2013
- TIA-568.2-D:2018

Standards for flammability, halogen acidity, and smoke performance

- IEC 60332-1-1:2015 (Ed. 1.1) / IEC 60332-1-2:2015 (Ed. 1.1)
- IEC 60754-2:2011 (Ed. 2.0)
- IEC 61034-1:2013 (Ed. 3.1) / IEC 61034-2:2013 (Ed. 3.1)

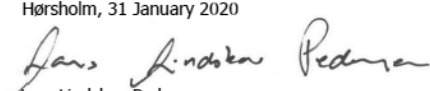
Technical report
T931777-02, DANAK-19/20916

EC Cabling product ID
5210

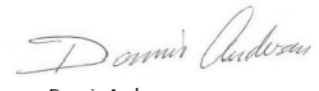
CS valid until
1 September 2020

This product has been tested by FORCE Technology and complies with the requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance schedule, which implies that FORCE Technology performs a quality audit of the manufacturer's production and QA sites. The maintenance testing of the product is performed on a sample basis.

Hørsholm, 31 January 2020



Lars Lindskov Pedersen
Test Manager



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Head of Department



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No. 2020-129

38 | LS Simple™ 지명원

www.lssimple.co.kr | 39



Compliance Statement

Data communication cable, Category 5e

Company
LS Cable & System Vietnam Co., Ltd.
Nhon Trach 2-Loc Khang Industrial Zone
Nhon Trach District
Dong Nai Province
Vietnam

Product description
Unscreened cable (U/UTP) characterised up to 100 MHz
Horizontal floor wiring cable, 100 Ω
4 twisted pairs
Flame retardant, PVC jacket

Product identification
Cat. 5e UTP PVC

Generic cabling and cabling components standards - Category 5e cable requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- IEC 61156-5:2012 (Ed. 2.1)
- EN 50173-1:2018 / EN 50173-2:2018
- EN 50288-3-1:2013
- TIA-568.2-D:2018

Standards for flame retardancy

- IEC 60332-1-1:2015 (Ed. 1.1)
- IEC 60332-1-2:2015 (Ed. 1.1)

Technical report
T931777-03, DANAK-19/20917

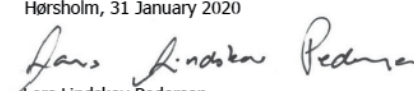
No. 2020-130

EC Cabling product ID
5005

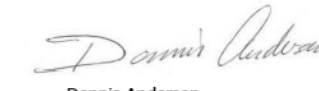
CS valid until
1 September 2020

This product has been tested by FORCE Technology and complies with the requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance schedule, which implies that FORCE Technology performs a quality audit of the manufacturer's production and QA sites. The maintenance testing of the product is performed on a sample basis.

Hørsholm, 31 January 2020



Lars Lindskov Pedersen
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Compliance Statement

Data communication cable, Category 6

Company
LS Cable & System Vietnam Co., Ltd.
Nhon Trach 2-Loc Khang Industrial Zone
Nhon Trach District
Dong Nai Province
Vietnam

Product description
Unscreened cable (U/UTP) characterised up to 250 MHz
Horizontal floor wiring cable, 100 Ω
4 twisted pairs
Flame retardant, PVC jacket

Product identification
Cat. 6 UTP PVC

Generic cabling and cabling components standards - Category 6 cable requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- IEC 61156-5:2012 (Ed. 2.1)
- EN 50173-1:2018 / EN 50173-2:2018
- EN 50288-6-1:2013
- TIA-568.2-D:2018

Standards for flame retardancy

- IEC 60332-1-1:2015 (Ed. 1.1)
- IEC 60332-1-2:2015 (Ed. 1.1)

Technical report
T931777-04, DANAK-19/20918

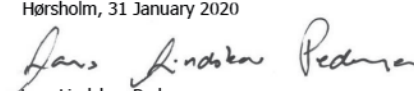
No. 2020-131

EC Cabling product ID
5211

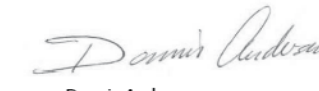
CS valid until
1 September 2020

This product has been tested by FORCE Technology and complies with the requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance schedule, which implies that FORCE Technology performs a quality audit of the manufacturer's production and QA sites. The maintenance testing of the product is performed on a sample basis.

Hørsholm, 31 January 2020



Lars Lindskov Pedersen
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Compliance Statement

No. 2020-132

Data communication cable, Category 7

Company
 LS Cable & System Vietnam Co., Ltd.
 Nhon Trach 2-Loc Khang Industrial Zone
 Nhon Trach District
 Dong Nai Province
 Vietnam

Product description
 Screened cable (F/FTP) characterised up to 600 MHz
 Horizontal floor wiring cable, 100 Ω
 4 individually foil screened twisted pairs with overall foil screen
 Flame retardant, halogen free

Product identification
 LS Cable & System, F/FTP Cat.7 LSZH
 Part Number: FSP-G-C7G-E1ZN-X 0.5x004P

Generic cabling and cabling components standards - Category 7 cable requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- IEC 61156-5:2012 (Ed. 2.1)
- EN 50173-1:2018 / EN 50173-2:2018
- EN 50288-4-1:2013

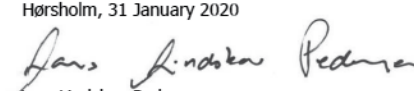
Standards for flammability, halogen acidity, and smoke performance

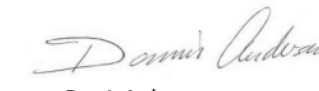
- IEC 60332-1-1:2015 (Ed. 1.1) / IEC 60332-1-2:2015 (Ed. 1.1)
- IEC 60754-2:2011 (Ed. 2.0)
- IEC 61034-1:2013 (Ed. 3.1) / IEC 61034-2:2013 (Ed. 3.1)

Technical report T926265-03, DANAK-19/20293	EC Cabling product ID 5723	CS valid until 1 September 2020
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This product has been tested by FORCE Technology and complies with the requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance schedule, which implies that FORCE Technology performs a quality audit of the manufacturer's production and QA sites. The maintenance testing of the product is performed on a sample basis.

Hørsholm, 31 January 2020


 Lars Lindskov Pedersen
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 Dennis Andersen
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Compliance Statement

No. 2019-680

Data communication cords, Category 6A

Company
 LS Cable & System Ltd.
 12~16F, LS Tower
 127, LS-ro, Dongan-gu
 Anyang-si, Gyeonggi-do
 Republic of Korea

Product description
 Screened Category 6A S/FTP LSZH RJ45 cords (equipment, patch, and work area) characterised up to 500 MHz

Product identification
 LS-SC6AL-XX-YYY
 XX denotes the color
 YYY denotes the length

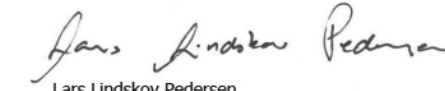
Generic cabling and cabling components standard - Category 6A balanced cord requirements

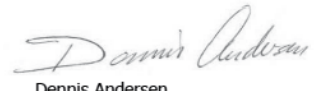
- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- EN 50173-1:2018 / EN 50173-2:2018
- TIA-568.2-D:2018
- IEC 61935-2:2010 (Ed. 3.0) (transmission requirements)

Technical report T933144-01, DANAK-19/20625	EC Cabling product ID 6066	CS valid until 1 October 2020
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This product has been tested by FORCE Technology and complies with the electrical requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance of certification schedule, which implies that FORCE Technology performs a sample test of the product once a year.

Hørsholm, 19 September 2019


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Compliance Statement

No. 2019-681

Data communication cords, Category 6A

Company
 LS Cable & System Ltd.
 12~16F, LS Tower
 127, LS-ro, Dongan-gu
 Anyang-si, Gyeonggi-do
 Republic of Korea

Product description
 Unscreened Category 6A U/UTP LSZH RJ45 (equipment, patch, and work area) characterised up to 500 MHz

Product identification
 LS-UC6AL-XX-YYY
 XX denotes the color
 YYY denotes the length

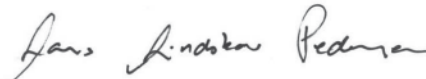
Generic cabling and cabling components standard - Category 6A balanced cord requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- EN 50173-1:2018 / EN 50173-2:2018
- TIA-568.2-D:2018
- IEC 61935-2:2010 (Ed. 3.0) (transmission requirements)

Technical report T933144-02, DANAK-19/20626	EC Cabling product ID 6067	CS valid until 1 October 2020
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This product has been tested by FORCE Technology and complies with the electrical requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance of certification schedule, which implies that FORCE Technology performs a sample test of the product once a year.

Hørsholm, 19 September 2019



Lars Lindskov Pedersen
Test Manager



Dennis Andersen
Head of Department



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Compliance Statement

No. 2019-682

Data communication cords, Category 6

Company
 LS Cable & System Ltd.
 12~16F, LS Tower
 127, LS-ro, Dongan-gu
 Anyang-si, Gyeonggi-do
 Republic of Korea

Product description
 Screened Category 6 S/FTP LSZH RJ45 cords (equipment, patch, and work area) characterised up to 250 MHz

Product identification
 LS-SC6L-XX-YYY
 XX denotes the color
 YYY denotes the length

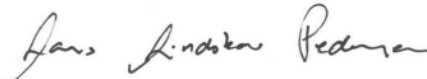
Generic cabling and cabling components standard - Category 6 balanced cord requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- EN 50173-1:2018 / EN 50173-2:2018
- TIA-568.2-D:2018
- IEC 61935-2:2010 (Ed. 3.0) (transmission requirements)

Technical report T933144-03, DANAK-19/20627	EC Cabling product ID 6068	CS valid until 1 October 2020
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This product has been tested by FORCE Technology and complies with the electrical requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance of certification schedule, which implies that FORCE Technology performs a sample test of the product once a year.

Hørsholm, 19 September 2019



Lars Lindskov Pedersen
Test Manager



Dennis Andersen
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Compliance Statement

No. 2019-683

Data communication cords, Category 6

Company

LS Cable & System Ltd.
12~16F, LS Tower
127, LS-ro, Dongan-gu
Anyang-si, Gyeonggi-do
Republic of Korea

Product description

Unscreened Category 6 U/UTP LSZH RJ45 cords (equipment, patch, and work area) characterised up to 250 MHz

Product identification

LS-UC6L-XX-YYY
XX denotes the color
YYY denotes the length

Generic cabling and cabling components standard - Category 6 balanced cord requirements

- ISO/IEC 11801-1:2017 (Ed. 1.0) / ISO/IEC 11801-2:2017 (Ed. 1.0)
- EN 50173-1:2018 / EN 50173-2:2018
- TIA-568.2-D:2018
- IEC 61935-2:2010 (Ed. 3.0) (transmission requirements)

Technical report

T933144-04, DANAK-19/20628

EC Cabling product ID

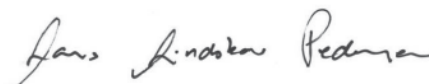
6069

CS valid until

1 October 2020

This product has been tested by FORCE Technology and complies with the electrical requirements of the above specified standards and "Terms and conditions for the EC VERIFIED programme for Generic and Coaxial Cabling", DQP231006. The product takes part in a maintenance of certification schedule, which implies that FORCE Technology performs a sample test of the product once a year.

Hørsholm, 19 September 2019



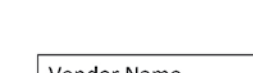
Lars Lindskov Pedersen
Test Manager



Dennis Andersen
Head of Department



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Vendor Name LS Cable & System Ltd.	
Cable model Name FTP-G-C6G-E1VN-M	
Device Vendor ID 0x113	
Product Category Cable	
Cable Type Cat.6 F/UTP 23AWG	Golden Units Firmware Version 7.3.14

Test ID	Title	Result	Non-Conformant Behavior Description

For HDBaseT Alliance Use Only	
Date Results Received	Date Vendor Application Received
Date Admin Fee Received	Date Recommendation Awarded
TSC Closed	Notice emailed to vendor Product posted on web

Page 5 of 12 CTS_Report_2.0c2_LS Cable & System_Cable_Cat6_FTP-G-C6G-E1VN-M_TSC_012018123101

전기안전인증 (멀티콘센트)

전기용품 및 생활용품 안전관리법 시행규칙 [별지 제5호서식]  접수번호 : 20180912-0017



안전인증서

Safety Certificate

안전인증번호: HU04158-18001A
(Certificate No.)
제조업자/수입업자명: NINGBO ZHENHAI JINZE ELECTRICAL APPLIANCE CO.,LTD
(Manufacturer/importer)
주 소: NO.3 BUILDING, BANLUWANG INDUSTRIAL ZONE, QINGSHUIPU VILLAGE,ZHENHAI DISTRICT, NINGBO CITY, ZHEJIANG
(Address)
제품명: 콘센트(이동형 멀티)
(Product)
기본모델명: LS-MC-S4-050
(Basic Model)
파생모델명 (Series Model):
LS-MC-S4-015 LS-MC-S4-030

정격/안전기준상의 모델구분: 16 A 250 V~
(Rating)
시험기준: KC 60884-1(2015-07) KS C 8305(2011-12)
(Standard)

본 인증서는 제조국명: 중국
제조업자명: NINGBO ZHENHAI JINZE ELECTRICAL APPLIANCE CO.,LTD
제조공장의 주소: NO.3 BUILDING, BANLUWANG INDUSTRIAL ZONE, QINGSHUIPU VILLAGE,ZHENHAI DISTRICT, NINGBO CITY, ZHEJIANG PROVINCE, CHINA
의 제품에만 해당함

「전기용품 및 생활용품 안전관리법 시행규칙」 제9조제2항, 제4항, 제10조제2항 또는 제15조제2항에 따라 안전인증서를 발급합니다.
We issue this Safety Certificate for the above appliances in accordance with the Article 9(2), 9(4), 10(2) or 15(2) of the Enforcement Rule of the Electrical Appliances and Consumer Products Safety Control Act.

2018 년 12 월 31 일
(Year) (Month) (Day)



한국기계전자시험연구원

Korea Testing Certification



※ 이 인증서는 「전기용품 및 생활용품 안전관리법」에 따른 제품의 안전성 확인에 한정된 것이며, 그 밖의 다른 법률이 적용되는 제품의 경우에는 해당 법률에 따라 추가로 인증·허가 등을 받아야 합니다.

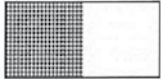
첨부서류	내용
1.	전기용품의 안전관리부품 및 재질목록 (List of Critical Components)
2.	기본모델·파생모델의 내용 (Descriptions of the basic and series model)
3.	안전인증의 변경 현황 (Revisions Status)



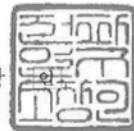
◎ 정보통신 기기인증

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정보통신기기인증서 (Cat.5E Patch Panel)

전자문서확인번호	MHNU-OFAD-KWP1-NU8N	
정보통신기기인증서		
Certificate of Information and Communication Equipment		
인 증 의 종 류 :	형식승인 (Type Approval)	
Certification Type		
상 호 또 는 성 명 :	엘에스전선 (주)	
Trade Name or Applicant		
기 기 의 명 칭 :	접속커넥터	
Equipment Name		
기 본 모 델 명 :	LS-PP-UC5E-24P	
Basic Model Number of Equipment		
파 생 모 델 명 :		
Series Model Number of Equipment		
인 증 번 호 :	LSC-PP-UC5E-24P	
Certification No		
제 조 자 및 제조국 가 :	HCI/대만	
Manufacturer and Country of Origin		
인 증 연 월 일 :	2006년 (Year) 11월 (Month) 20일 (Date)	
Date of Certification		
기 타 :		
Others		
위 기기는 정보통신기기 인증규칙에 의해 인증되었음을 증명합니다. It is certified that foregoing equipment has been certificated under the provisions of the Regulations on Certification of Information and Communication Equipment.		
2008년 (Year) 04월 (Month) 08일 (Date) 전파연구소장		
Director General of Radio Research Laboratory Broadcasting and Communications Commission Republic of Korea		
※ 진위여부는 www.ebcc.go.kr에서 확인할 수 있습니다. ※ 복사본은 상단 복사방지마크의 '원' 또는 '본' 글자가 사라집니다.		

정보통신기기인증서 (Cat.6 Patch Panel)

전자문서확인번호	TQY1-YD12-F11K-1N01	
정보통신기기인증서		
Certificate of Information and Communication Equipment		
인 증 의 종 류 :	형식승인 (Type Approval)	
Certification Type		
상 호 또 는 성 명 :	엘에스전선 (주)	
Trade Name or Applicant		
기 기 의 명 칭 :	접속커넥터	
Equipment Name		
기 본 모 델 명 :	LS-PP-UC6-24P-WM	
Basic Model Number of Equipment		
파 생 모 델 명 :		
Series Model Number of Equipment		
인 증 번 호 :	LSC-PP-UC6-24P-WM	
Certification No		
제 조 자 및 제조국 가 :	HCI/대만	
Manufacturer and Country of Origin		
인 증 연 월 일 :	2006년 (Year) 11월 (Month) 17일 (Date)	
Date of Certification		
기 타 :		
Others		
위 기기는 정보통신기기 인증규칙에 의해 인증되었음을 증명합니다. It is certified that foregoing equipment has been certificated under the provisions of the Regulations on Certification of Information and Communication Equipment.		
2008년 (Year) 04월 (Month) 08일 (Date) 전파연구소장		
Director General of Radio Research Laboratory Broadcasting and Communications Commission Republic of Korea		
※ 진위여부는 www.ebcc.go.kr에서 확인할 수 있습니다. ※ 복사본은 상단 복사방지마크의 '원' 또는 '본' 글자가 사라집니다.		

정보통신기기인증서 (Modular Jack 종합)

정보통신기기인증서 (LED Modular Jack)

8C95-16A2-9D1A-47F7

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MJ-UCC6-WH-ERI
파생모델명 Series Model Number	LS-MJ-UC5E-BK-ERI, LS-MJ-UC6A-GY-ERI, LS-MJ-UC6A-RD-ERI, LS-MJ-UC6A-BK-ERI, LS-MJ-UCC6-YL-ERI, LS-MJ-UCC6-GN-ERI, LS-MJ-UCC6-BL-ERI, LS-MJ-UC5E-YL-ERI, LS-MJ-UC5E-GN-ERI, LS-MJ-UC5E-BL-ERI, LS-MJ-UC5E-WH-ERI, LS-MJ-UC6A-YL-ERI, LS-MJ-UC6A-GN-ERI, LS-MJ-UC6A-BL-ERI, LS-MJ-UC6A-WH-ERI, LS-MJ-UCC6-GY-ERI, LS-MJ-UCC6-RD-ERI, LS-MJ-UCC6-BK-ERI, LS-MJ-UC5E-GY-ERI, LS-MJ-UC5E-RD-ERI
등록번호 Registration No.	R-RTI-LS1-MJ-UCXX-XX-ERI
제조사/제조(조립)국가 Manufacturer/Country of Origin	Rubiconn. Inc / 대만
등록연월일 Date of Registration	2018-06-05
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2018년(Year) 06월(Month) 05일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

57B5-E21C-ABAD-8496

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MJ-UCC6-WH-SRI
파생모델명 Series Model Number	LS-MJ-UCC6-BL-SRI, LS-MJ-UC5E-YL-SRI, LS-MJ-UC5E-WH-SRI, LS-MJ-UC5E-RD-SRI, LS-MJ-UC5E-BK-SRI, LS-MJ-UC5E-GN-SRI, LS-MJ-UC5E-GY-SRI, LS-MJ-UC5E-BL-SRI, LS-MJ-UC6A-YL-SRI, LS-MJ-UC6A-GN-SRI, LS-MJ-UC6A-GY-SRI, LS-MJ-UC6A-RD-SRI, LS-MJ-UC6A-WH-SRI, LS-MJ-UCC6-YL-SRI, LS-MJ-UCC6-GN-SRI, LS-MJ-UCC6-RD-SRI, LS-MJ-UCC6-BK-SRI, LS-MJ-UC6A-BL-SRI, LS-MJ-UC6A-BK-SRI, LS-MJ-UCC6-GY-SRI
등록번호 Registration No.	R-RTI-LS1-MJ-UCXX-XX-SRI
제조사/제조(조립)국가 Manufacturer/Country of Origin	Rubiconn. Inc / 대만
등록연월일 Date of Registration	2018-06-05
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2018년(Year) 08월(Month) 20일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

정보통신기기인증서 (Cat.6 비차폐 Modular Plug)

정보통신기기인증서 (Cat.6 차폐 EZ Modular Plug)

2A92-4561-E8ED-E78C

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MP-UC6-RJ45-TR-A1
파생모델명 Series Model Number	LS-MP-UC6(A)-RJ45B-GK, LS-MP-UC6A-RJ45-TR-A7, LS-MP-UC6-RJ45-TR-A6, LS-MP-UC6-RJ45-EZ, LS-MP-UC6(A)-RJ45B-K
등록번호 Registration No.	MSIP-RTI-LS1-MP1-UC6-RJ45
제조사/제조(조립)국가 Manufacturer/Country of Origin	COBTTEL PRECISION ELETRONICS CO.,LTD / 중국
등록연월일 Date of Registration	2017-04-10
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2019년(Year) 01월(Month) 21일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

CA8D-749D-2C74-CBA7

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MP-SC6-RJ45-EZ-C
파생모델명 Series Model Number	
등록번호 Registration No.	R-R-LS1-MP1-SC-EZ
제조사/제조(조립)국가 Manufacturer/Country of Origin	COBTTEL PRECISION ELETRONICS CO., LTD. / 중국
등록연월일 Date of Registration	2019-12-17
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2019년(Year) 12월(Month) 17일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

정보통신기기인증서 (Cat.6 비차폐 EZ Modular Plug)

정보통신기기인증서 (Cat.5E Modular Plug)

E43B-A461-3AAE-21D1

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MP-UC6-RJ45-EZ-C
파생모델명 Series Model Number	
등록번호 Registration No.	R-R-LS1-MP1-UC6-EZ
제조사/제조(조립)국가 Manufacturer/Country of Origin	COBTEL PRECISION ELETRONICS CO., LTD. / 중국
등록연월일 Date of Registration	2019-12-17
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2019년(Year) 12월(Month) 17일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

240A-5C7B-45D9-3525

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MP-UC5E-RJ45
파생모델명 Series Model Number	LS-MP-UC5E-RJ45B, LS-MP-UC5E-RJ45-EZK-RD, LS-MP-UC5E-RJ45-EZK-BL, LS-MP-UC5E-RJ45-EZ, LS-MP-UC5E-RJ45-EZK-TR
등록번호 Registration No.	MSIP-RTI-LS1-MP1-UC-RJ45
제조사/제조(조립)국가 Manufacturer/Country of Origin	COBTEL PRECISION ELETRONICS CO.,LTD. / 중국
등록연월일 Date of Registration	2017-02-20
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2018년(Year) 10월(Month) 29일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

6E8E-416E-89B4-50A0

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재명칭(제품명칭) Equipment Name	접속커넥터
기본모델명 Basic Model Number	LS-MP-SC-RJ45-TR-D1
파생모델명 Series Model Number	LS-MP-SC-RJ45-TR-B2, LS-MP-SC-RJ45-TR-D5, LS-MP-SC6(A)-RJ45B-K, LS-MP-SC6(A)-RJ45B-GK, LS-MP-SC-RJ45-TR-B1
등록번호 Registration No.	MSIP-RTI-LS1-MP1-SC-RJ45
제조사/제조(조립)국가 Manufacturer/Country of Origin	COBTTEL PRECISION ELETRONICS CO.,LTD. / 중국
등록연월일 Date of Registration	2017-01-04
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2018년(Year) 05월(Month) 21일(Day)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

25ED-CB3D-72C0-3515

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재 명칭 Equipment Name	동축케이블
기본모델명 Basic Model Number	5C-HFBT
파생모델명 Series Model Number	
등록번호 Registration No.	MSIP-RTS-LS1-5C-HFBT
제조사/제조(조립)국가 Manufacturer/Country of Origin	알엔케이 (주) / 한국
등록연월일 Date of Registration	2014-03-28
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2014년(Year) 03월(Month) 28일(Date)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

정보통신기기인증서 (HFBT 7C 60% AL)

9563-7314-7138-F5F9

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재 명칭 Equipment Name	동축케이블
기본모델명 Basic Model Number	7C-HFBT
과생모델명 Series Model Number	
등록번호 Registration No.	MSIP-RTS-LS1-7C-HFBT
제조사/제조(조립)국가 Manufacturer/Country of Origin	알엔케이블 (주) / 한국
등록연월일 Date of Registration	2014-07-03
기타 Others	
위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다. It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act. 2014년(Year) 07월(Month) 03일(Date) 국립전파연구원장 Director General of National Radio Research Agency ※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다. 위반시 과태료 처분 및 등록이 취소될 수 있습니다.	

정보통신기기인증서 (HFBT 10C 60% AL)

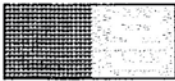
B317-2D89-9A02-CE15

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재 명칭 Equipment Name	동축케이블
기본모델명 Basic Model Number	10C-HFBT
과생모델명 Series Model Number	
등록번호 Registration No.	MSIP-RTS-LS1-10C-HFBT
제조사/제조(조립)국가 Manufacturer/Country of Origin	알엔케이블 (주) / 한국
등록연월일 Date of Registration	2014-03-28
기타 Others	
위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다. It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act. 2014년(Year) 03월(Month) 28일(Date) 국립전파연구원장 Director General of National Radio Research Agency ※ 적합등록 방송통신기자재는 반드시 "적합성평가표시" 를 부착하여 유통하여야 합니다. 위반시 과태료 처분 및 등록이 취소될 수 있습니다.	

정보통신기기인증서 (SimpleWin Controller)

8A24-EBF9-39DB-7427

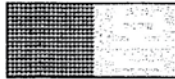
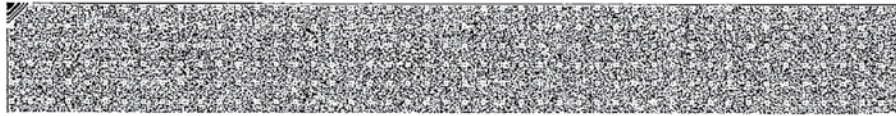
방송통신기자재등의 적합등록 필증 <i>Registration of Broadcasting and Communication Equipments</i>	
상호 또는 성명 <i>Trade Name or Registrant</i>	엘에스전선(주)
기자재 명칭 <i>Equipment Name</i>	Simplewin 컨트롤러 대형
기본모델명 <i>Basic Model Number</i>	LS-IC960-PS2A
파생모델명 <i>Series Model Number</i>	LS-IC960-PS2S,LS-IC960-PS1S,LS-IC960-PS1A
등록번호 <i>Registration No.</i>	MSIP-REM-LS1-LS-IC960-PS2A
제조사/제조(조립)국가 <i>Manufacturer/Country of Origin</i>	엘에스전선(주) / 한국
등록연월일 <i>Date of Registration</i>	2016-07-27
기타 <i>Others</i>	
위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다. It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act. 2016년(Year) 07월(Month) 27일(Date) 국립전파연구원장 <i>Director General of National Radio Research Agency</i> ※ 적합등록 방송통신기자재는 반드시 "적합성평가표시"를 부착하여 유통하여야 합니다. 위반시 과태료 처분 및 등록이 취소될 수 있습니다.	




정보통신기기인증서 (SimpleWin Patch Panel)

6075-1547-EC18-995A

방송통신기자재등의 적합등록 필증 <i>Registration of Broadcasting and Communication Equipments</i>	
상호 또는 성명 <i>Trade Name or Registrant</i>	엘에스전선(주)
기자재 명칭 <i>Equipment Name</i>	Simplewin 컨트롤러 중형
기본모델명 <i>Basic Model Number</i>	LS-IC384-PS1A
파생모델명 <i>Series Model Number</i>	LS-IC384-PS1S
등록번호 <i>Registration No.</i>	MSIP-REM-LS1-LS-IC384-PS1A
제조사/제조(조립)국가 <i>Manufacturer/Country of Origin</i>	엘에스전선(주) / 한국
등록연월일 <i>Date of Registration</i>	2016-07-27
기타 <i>Others</i>	
위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다. It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act. 2016년(Year) 07월(Month) 27일(Date) 국립전파연구원장 <i>Director General of National Radio Research Agency</i> ※ 적합등록 방송통신기자재는 반드시 "적합성평가표시"를 부착하여 유통하여야 합니다. 위반시 과태료 처분 및 등록이 취소될 수 있습니다.	

4A8B-2E85-5A9A-BD4C

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재 명칭 Equipment Name	Simplewin 패치패널
기본모델명 Basic Model Number	LS-IPPC-UC-24P
파생모델명 Series Model Number	LS-IPPI-UC-24P,LS-IAPC-UC-24P,LS-IFDC-LC-SM-C-048-DP-D,LS-IFDC-LC-MM-C-048-DP-D,LS-IAPC-UC-24P,LS-IAPC-SC-24P,LS-IFDC-LC-M3-C-048-DP-D,LS-IFDC-LC/MPO-SM-C-048-DP-D,LS-IFDC-LC/MPO-M3-C-048-DP-D,LS-IFDC-LC-M4-C-048-DP-D,LS-IFDC-LC/MPO-MM-C-048-DP-D,LS-IFDC-LC/MPO-M4-C-048-DP-D,LS-IPPC-SC-24P
등록번호 Registration No.	MSIP-REM-LS1-LS-IPPC-UC-24P
제조사/제조(조립)국가 Manufacturer/Country of Origin	엘에스전선(주) / 한국
등록연월일 Date of Registration	2016-07-27
기타 Others	
위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다. It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act. 2016년(Year) 08월(Month) 09일(Date) 국립전파연구원장 Director General of National Radio Research Agency ※ 적합등록 방송통신기자재는 반드시 "적합성평가표시"를 부착하여 유통하여야 합니다. 위반시 과태료 처분 및 등록이 취소될 수 있습니다.	

Korea Testing Laboratory

1등급
GOOD Software

소프트웨어품질인증서 Certificate of software Quality

상 호 또 는 성 명 : 엘에스전선(주)
Name of Company/Name of Applicant : LS Cable & System Ltd.

소 프 트 웨 어 의 명 칭 : SimpleWin v2.0
Name of Software : 심플윈 v2.0

인 증 등 급 : 1 등급 (소프트웨어품질인증의 최고등급)
Certification Level : Level1 (The Highest level of Software Quality Certification)

인 증 번 호 : 16-0017
Certification No.

제 조 자 및 제 조 국 가 : 엘에스전선(주) / 대한민국
Manufacturer and Country of Manufacture : LS Cable & System Ltd. / Republic of Korea

인 증 연 월 일 : 2016년(Year) 10월(Month) 27일(Day)
Date of certification

기 타 :
Additional Information :
 ■ 인증 품목명 : LS-IC384-PS1A 용 통신배선 관리 시스템
 ■ 제품 구성 모듈 : SimpleWin v2.0 Server for Windows, SimpleWin v2.0 Client for Windows

위 소프트웨어는 소프트웨어산업진흥법 제13조 및 동법 시행규칙 제5조 제1항에 따라 인증되었음을 증명합니다.

I hereby confirm that the quality of the foregoing software has been certified under Article 13 of the Software Industry Promotion Act, and Article 5.1 of the Enforcement Decree of the Software Industry Promotion Act.

2016년(Year) 10월(Month) 27일(Day)

ktl 한국산업기술시험원장
Korea Testing Laboratory

President of the Software Quality Certification Body Republic of Korea

적합등록필증 (관로용 광케이블)

2D7E-BD6A-A9EC-8224

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재 명칭 Equipment Name	관로용
기본모델명 Basic Model Number	LT-MJBAEXX/BRN SE288X(12T2.30)
파생모델명 Series Model Number	광전복합, SJNA, DJSA, Central Tube, SJSA
등록번호 Registration No.	MSIP-RTM-LS1-LT-MJBAEXX-BRN
제조사/제조(조립)국가 Manufacturer/Country of Origin	엘에스전선(주) / 한국
등록연월일 Date of Registration	2015-03-23
기타 Others	

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It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2015년(Year) 07월(Month) 15일(Date)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시"를 부착하여 유통하여야 합니다.
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적합등록필증 (루즈튜브 광케이블)

5254-7B57-9D48-EE0D

방송통신기자재등의 적합등록 필증 Registration of Broadcasting and Communication Equipments	
상호 또는 성명 Trade Name or Registrant	엘에스전선(주)
기자재 명칭 Equipment Name	광케이블 Loose tube
기본모델명 Basic Model Number	Loose tube type
파생모델명 Series Model Number	SJSA, TJSA, 광전복합, Central Tube, Fig-8, DJSA, SJNA
등록번호 Registration No.	MSIP-RTM-LS1-LT-CABLE
제조사/제조(조립)국가 Manufacturer/Country of Origin	엘에스전선(주) / 한국
등록연월일 Date of Registration	2016-03-30
기타 Others	

위 기자재는 「전파법」 제58조의2 제3항에 따라 등록되었음을 증명합니다.
It is verified that foregoing equipment has been registered under the Clause 3, Article 58-2 of Radio Waves Act.

2016년(Year) 03월(Month) 30일(Date)

국립전파연구원장
Director General of National Radio Research Agency

※ 적합등록 방송통신기자재는 반드시 "적합성평가표시"를 부착하여 유통하여야 합니다.
위반시 과태료 처분 및 등록이 취소될 수 있습니다.

◎ 선급인증

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DNV·GL

Certificate No:
TAE000035R
Revision No:
1

TYPE APPROVAL CERTIFICATE

This is to certify:

That the Category cables

with type designation(s)
FTP Cat 6 LSZH, F/UTP Cat 6A LSZH

Issued to
LS Cable & System Vietnam
Dong Nai, Viet Nam

is found to comply with
DNV GL rules for classification – Ships, offshore units, and high speed and light craft

Application :

Product(s) approved by this certificate is/are accepted for installation on all vessels classed by DNV GL.

Issued at **Hamburg** on **2019-09-17**

This Certificate is valid until **2023-09-13**.
DNV GL local station: **Vung Tau Station**

Approval Engineer: **Carsten Hunsalz**

for **DNV GL**
Arne Schaarmann
Head of Section

This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid. The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.



Form code: TA 251

Revision: 2016-12

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Page 1 of 4

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DNV·GL

Certificate No:
TAE0000341

TYPE APPROVAL CERTIFICATE

This is to certify:

That the Category cables

with type designation(s)
4 PAIR S/FTP Cat 5e LSZH, 4 PAIR S/FTP Cat 6/6A LSZH, 4 PAIR S/FTP Cat 7 LSZH

Issued to
LS Cable & System Vietnam
Dong Nai, Viet Nam

is found to comply with
DNV GL rules for classification – Ships, offshore units, and high speed and light craft

Application :

Product(s) approved by this certificate is/are accepted for installation on all vessels classed by DNV GL.

Issued at **Hamburg** on **2018-09-14**

This Certificate is valid until **2023-09-13**.
DNV GL local station: **Vung Tau Station**

Approval Engineer: **Carsten Hunsalz**

for **DNV GL**
Arne Schaarmann
Head of Section

This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid. The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.



Form code: TA 251

Revision: 2016-12

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		DNV GL	
TYPE APPROVAL CERTIFICATE		Certificate No: TAE0000139 Revision No: 1	
This is to certify:			
That the Category cables			
with type designation(s) SF-UTP Cat 5e LSZH			
Issued to LS Cable & System Vietnam Dong Nai, Viet Nam			
is found to comply with DNV GL rules for classification – Ships, offshore units, and high speed and light craft			
Application :			
Product(s) approved by this certificate is/are accepted for installation on all vessels classed by DNV GL.			
Issued at Hamburg on 2018-09-14		for DNV GL	
This Certificate is valid until 2023-09-13 .			
DNV GL local station: Vung Tau Station			
Approval Engineer: Carsten Hunsalz		Arne Schaarmann Head of Section	
This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid. The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.			
	Form code: TA 251	Revision: 2016-12	www.dnvgl.com
		Page 1 of 3	
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Certificate Number: 19-SG1844751-PDA 16/JUN/2019	
	
Confirmation of Product Type Approval	
Please refer to the "Service Restrictions" shown below to determine if Unit Certification is required for this product. This certificate reflects the information on the product in the ABS Records as of the date and time the certificate is printed. Pursuant to the Rules of the American Bureau of Shipping (ABS), the manufacturer of the below listed product held a valid Manufacturing Assessment (MA) with expiration date of 05-MAY-2024. The continued validity of the Manufacturing Assessment is dependent on completion of satisfactory audits as required by the ABS Rules. And; a Product Design Assessment (PDA) valid until subject to continued compliance with the Rules or standards used in the evaluation of the product. The above entitle the product to be called Product Type Approved. The Product Design Assessment is valid for products intended for use on ABS classed vessels, MODUs or facilities which are in existence or under contract for construction on the date of the ABS Rules used to evaluate the Product. ABS makes no representations regarding Type Approval of the Product for use on vessels, MODUs or facilities built after the date of the ABS Rules used for this evaluation. Due to wide variety of specifications used in the products ABS has evaluated for Type Approval, it is part of our contract that; whether the standard is an ABS Rule or a non-ABS Rule, the Client has full responsibility for continued compliance with the standard.	
Product Name: Communication Cable	
Model Name(s): SF/UTP Category 5e LSZH (CMX); FTP Category 6 LSZH (CMX)	
Presented to: LS CABLE & SYSTEM VIETNAM CO., LTD. NHONTRACH II-LOC KHANG INDUSTRIAL ZONE HIEP PHUOC VILLAGE, NHON TRACH DISTRICT Vietnam	
Intended Service:	Communication cable for Marine Shipboard and Offshore application
Description:	Screened foiled twisted pair (SF/UTP) cables of 100Ohms, Enhanced Category 5 (Cat.5E). Foiled twisted pair (F/UTP) cables of 100Ohms, Category 6 (Cat.6)
Tier:	3
Ratings:	SF/UTP Cat.5E Conductor: Solid, annealed and bare copper Insulation: High Density Polyethylene Core Assembly: Four twisted pairs Screen: Aluminum tape coated on one side with plastic film shall be applied over the core wrapping for screening. The braided screen with tinned copper wires shall be applied over aluminum tape screen. Sheath: Flame retardant PVC or LSZH (Low smoke Zero Halogen) FTP Cat.6 Conductor: Solid, annealed and bare copper Insulation: High Density Polyethylene Core Assembly: Four twisted pairs, The cross shape filler inserted between pairs. Screen: Aluminum tape coated on one side with plastic film shall be applied over the cable core for screening on jacketing process. The tinned copper wire with the diameter of 0.4mm shall be applied with cable core in jacket. Sheath: Flame retardant PVC or LSZH (Low smoke Zero Halogen)
Service Restrictions:	1. Unit Certification is not required for this product. 2. If the manufacturer or purchaser request an ABS Certificate for compliance with a specification or standard, the specification or standard, including inspection standards and tolerances, must be clearly defined.

ABS (SF/UTP Cat.5E LSZH, F/UTP Cat.6 LSZH)

ABS (S/FTP Cat.6A Dual Jacket)

Comments:	1. The Manufacturer has provided a declaration about the control of, or the lack of Asbestos in this product. 2. Each particular application and installation is to be specifically approved in conjunction with the relevant system, locations and environmental conditions.		
Notes / Documentation:	Test Report No: 1809-001_FTP Cat.6 LSZH test report_ABS – “Communication cable (FTP, category 6, 24 AWG, 4 Pairs, LSZH Jacket” carried out by LS Cable & System Vietnam Co, Ltd dated 10 Sep 2018. Test Report No: 1809-002_SF-UTP Cat.5e LSZH test report_ABS – “Communication cable (SF-UTP Category 5e, 24 AWG, 4 Pairs, LSZH Jacket” carried out by LS Cable & System Vietnam Co, Ltd dated 10 Sep 2018. Test Report No: T2018-12289 “Flame requirement, Smoke density, Acidity and Conductivity test” carried out by Korea Testing Certification dated 24 Oct 2018 to 4 Dec 2018.		
Term of Validity:	This Product Design Assessment (PDA) Certificate 19-SG1844751-PDA, dated 06/May/2019 remains valid until 05/May/2024 or until the Rules or specifications used in the assessment are revised (whichever occurs first). This PDA is intended for a product to be installed on an ABS classed vessel, MODU or facility which is in existence or under contract for construction on the date of the ABS Rules or specifications used to evaluate the Product. Use of the Product on an ABS classed vessel, MODU or facility which is contracted after the validity date of the ABS Rules and specifications used to evaluate the Product, will require re-evaluation of the PDA. Use of the Product for non ABS classed vessels, MODUs or facilities is to be to an agreement between the manufacturer and intended client.		
ABS Rules:	Rules for Conditions of Classification (2019) – Steel Vessels 1-1-4/7.7, 1-1-A3, 1-1-A4, which covers the following: 2019 Rules for Building and Classing Marine Vessels: 4-8-3/9.1, 4-8-3/9.5 & 4-8-3/9.13 2019 Rules for Building and Classing Steel Vessels: 4-8-3/9.1, 4-8-3/9.5 & 4-8-3/9.13 2019 Rules for Building and Classing Offshore Support Vessels (OSV Rules): 4-8-3/9.1, 4-8-3/9.5 & 4-8-3/9.13 2019 Rules for Building and Classing Steel Vessels Under 90m in Length: 4-6-3/5.21, 4-6-4/13.1.1, 4-6-4/13.1.2 ABS Rules for Conditions of Classification (2019) – High Speed Craft 1-1-4/7.7, 1-1-A2, 1-1-A3, which covers the following: 2019 Rules for Building and Classing High Speed Craft: 4-6-3/5.21, 4-6-4/13.1.1, 4-6-4/13.1.2 ABS Rules for Conditions of Classification (2019) – Offshore Units and Structures 1-1-4/7.7, 1-1-A2 and A3, which covers the following: 2019 Mobile Offshore Drilling Units: 4-3-4/7.1.1, 4-3-4/7.1.2, 4-3-4/7.1.6		
National Standards:			
International Standards:	IEC 61156-5 Ed 2.1 en: 2012 IEC 60754-2 Ed 2.0 B:2011 IEC 61034-2 Ed 3.1 B:2013 IEC 60332-1-2 Ed.1.1:2015		
Government Authority:			
EUMED:			
Others:			
Model Certificate	Model Certificate No	Issue Date	Expiry Date
PDA	19-SG1844751-PDA	06-MAY-2019	05-MAY-2024
 ABS Programs ABS has used due diligence in the preparation of this certificate and it represents the information on the product in the ABS Records as of the date and time the certificate was printed. Type Approval requires Drawing Assessment, Prototype Testing and assessment of the manufacturer's quality assurance and quality control arrangements. Limited circumstances may allow only Prototype Testing to satisfy Type Approval. The approvals of Drawings and Products remain valid as long as the ABS Rule, to which they were assessed, remains valid. ABS cautions manufacturers to review and maintain compliance with all other specifications to which the product may have been assessed. Further, unless it is specifically indicated in the description of the product; Type Approval does not necessarily waive witnessed inspection or survey procedures (where otherwise required) for products to be used in a vessel, MODU or facility intended to be ABS classed or that is presently in class with ABS. Questions regarding the validity of ABS Rules or the need for supplemental testing or inspection of such products should, in all cases, be addressed to ABS.			

06/16/2019 11:23:26 PM

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Page 2 of 2

Certificate Number: 15-SG1441980-PDA 18/NOV/2015															
 Confirmation of Product Type Approval Please refer to the "Service Restrictions" shown below to determine if Unit Certification is required for this product. This certificate reflects the information on the product in the ABS Records as of the date and time the certificate is printed. Pursuant to the Rules of the American Bureau of Shipping (ABS), the manufacturer of the below listed product held a valid Manufacturing Assessment (MA) with expiration date of 26/DEC/2018. The continued validity of the Manufacturing Assessment is dependent on completion of satisfactory audits as required by the ABS Rules. And; a Product Design Assessment (PDA) valid until 26/OCT/2020 subject to continued compliance with the Rules or standards used in the evaluation of the product. The above entitle the product to be called Product Type Approved. The Product Design Assessment is valid for products intended for use on ABS classed vessels, MODUs or facilities which are in existence or under contract for construction on the date of the ABS Rules used to evaluate the Product. ABS makes no representations regarding Type Approval of the Product for use on vessels, MODUs or facilities built after the date of the ABS Rules used for this evaluation. Due to wide variety of specifications used in the products ABS has evaluated for Type Approval, it is part of our contract that; whether the standard is an ABS Rule or a non-ABS Rule, the Client has full responsibility for continued compliance with the standard. Product Name: Cable, Communication Model Name(s): 4 Pair S/FTP Dual Jacket Cables (500MHz Category 6A) Presented to: LS CABLE & SYSTEM VIETNAM CO., LTD. NHONTRACH II-LOC KHANG INDUSTRIAL ZONE HIEP PHUOC VILLAGE, NHON TRACH DISTRICT Vietnam <table><tr><td>Intended Service:</td><td>Marine and Offshore Application – Control, Instrumentation and Communication Systems</td></tr><tr><td>Description:</td><td>Screened cable (S/FTP) characterized up to 500MHz Horizontal floor wiring cable, 100# 4 individually foil screened twisted pairs with overall braid screen Flame retardant, halogen free, dual jacket Each conductor shall be insulated skin foam skin 3 layers with polyethylene</td></tr><tr><td>Tier:</td><td>3</td></tr><tr><td>Ratings:</td><td>Voltage: 72VDC, Temperature: 75°C</td></tr><tr><td>Service Restrictions:</td><td>Unit certification is required for propulsion system cables and thruster cables intended for Dynamic Positioning</td></tr><tr><td>Comments:</td><td>The Manufacturer has provided a declaration about the control of, or the lack of Asbestos in this product.</td></tr><tr><td>Notes / Documentation:</td><td>Drawing No. ABS Witness Review, ABS Witness Review, Revision: A, Pages: 8 Drawing No. DCM Test Report_Alien Cross-talk test, DCM Test Report_Alien Cross-talk test, Revision: A, Pages: 5 Drawing No. DCM Test Report_Electrical Performance Test, DCM Test report_Electrical Performance Test, Revision: A, Pages: 5 Drawing No. DCM Test Report_TCL-ELTCTL Test, DCM Test Report_TCL-ELTCTL Test, Revision: A, Pages: 5 Drawing No. Delta Test report_T211490 LS Cable Vietnam, Delta Test report_T211490 LS Cable Vietnam,</td></tr></table>		Intended Service:	Marine and Offshore Application – Control, Instrumentation and Communication Systems	Description:	Screened cable (S/FTP) characterized up to 500MHz Horizontal floor wiring cable, 100# 4 individually foil screened twisted pairs with overall braid screen Flame retardant, halogen free, dual jacket Each conductor shall be insulated skin foam skin 3 layers with polyethylene	Tier:	3	Ratings:	Voltage: 72VDC, Temperature: 75°C	Service Restrictions:	Unit certification is required for propulsion system cables and thruster cables intended for Dynamic Positioning	Comments:	The Manufacturer has provided a declaration about the control of, or the lack of Asbestos in this product.	Notes / Documentation:	Drawing No. ABS Witness Review, ABS Witness Review, Revision: A, Pages: 8 Drawing No. DCM Test Report_Alien Cross-talk test, DCM Test Report_Alien Cross-talk test, Revision: A, Pages: 5 Drawing No. DCM Test Report_Electrical Performance Test, DCM Test report_Electrical Performance Test, Revision: A, Pages: 5 Drawing No. DCM Test Report_TCL-ELTCTL Test, DCM Test Report_TCL-ELTCTL Test, Revision: A, Pages: 5 Drawing No. Delta Test report_T211490 LS Cable Vietnam, Delta Test report_T211490 LS Cable Vietnam,
Intended Service:	Marine and Offshore Application – Control, Instrumentation and Communication Systems														
Description:	Screened cable (S/FTP) characterized up to 500MHz Horizontal floor wiring cable, 100# 4 individually foil screened twisted pairs with overall braid screen Flame retardant, halogen free, dual jacket Each conductor shall be insulated skin foam skin 3 layers with polyethylene														
Tier:	3														
Ratings:	Voltage: 72VDC, Temperature: 75°C														
Service Restrictions:	Unit certification is required for propulsion system cables and thruster cables intended for Dynamic Positioning														
Comments:	The Manufacturer has provided a declaration about the control of, or the lack of Asbestos in this product.														
Notes / Documentation:	Drawing No. ABS Witness Review, ABS Witness Review, Revision: A, Pages: 8 Drawing No. DCM Test Report_Alien Cross-talk test, DCM Test Report_Alien Cross-talk test, Revision: A, Pages: 5 Drawing No. DCM Test Report_Electrical Performance Test, DCM Test report_Electrical Performance Test, Revision: A, Pages: 5 Drawing No. DCM Test Report_TCL-ELTCTL Test, DCM Test Report_TCL-ELTCTL Test, Revision: A, Pages: 5 Drawing No. Delta Test report_T211490 LS Cable Vietnam, Delta Test report_T211490 LS Cable Vietnam,														

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Page 1 of 2



LS CABLE & SYSTEM LTD – INDONG PLANT
643, JINPYUNG-DONG, GUMI-SI,
KOREA

Reference: YKP/kt/T1511729
Project Number: 2257355
Date: 11 May 2016
Page: 1 / 4

ATTN : Mr. B.J. Park / Manager of Telecommunication Quality Assurance

Review Activity : Type Approval Design Assessment (PDA) for:
Cable Spec. No. LSSS-LN0140-02 (4 Pair S/FTP Galvanized Steel Wire Braid
Armored Cables (PiMF 500MHz Category 6A)
Your Ref.: Letter dated 04 January 2016 and Email dated 25 January 2016 & 14
April 2016

Gentlemen:

We have received your declaration stating that the materials in the requested assessment are free from Asbestos and your application for the ABS Type Approval dated 04 January 2016, 25 January 2016 and 14 April 2016 together with the enclosures listed below. Having reviewed the enclosures in accordance with the following Standards/Codes,

- 2016 Steel Vessels Rules 1-1-4/7.7, 1-1/A3, 1-1/A4, 4-1-1/Table3, 4-8-3/9,
- 2016 Steel Vessels Under 90 Meters (295 Feet) in Length 1-1-4/7.7, 1-1/A3, 1-1/A4, 4-1-1/Table3C, 4-6-3/5,
- 2016 Offshore Support Vessels 1-1-4/7.7, 1-1/A3, 1-1/A4, 4-1-1/Table3, 4-8-3/9,
- 2016 Facilities on Offshore Installations 1-1-4/9.7, 1-1/A2, 1-1/A3,
- 2016 Mobile Offshore Drilling Units Rules 1-1-4/9.7, 1-1/A2, 1-1/A3, 4-3-4/7.1
- UL 1581 (2008), IEC Publications 61156-5 (2012), 61034-2 (2013), 60754-2 (2011), 60332-3-22 (2009)

We are pleased to advise that we have completed the Product Design Assessment (PDA) phase of the type approval process. Enclosed is your original copy of the PDA. Your details are published on our web site at www.typeapproval.org and can also be downloaded there.

Please be sure that you understand the scope and conditions of the validity of the certificate. An assessment is not an approval. As applicable, the PDA is a generic assessment of materials, components, products or systems for a specific use in compliance with the Rules, Guides and recognized standards. In essence the final approval is given when an Engineer or Surveyor accepts it for a specific user and installation. Therefore, the PDAs are written assuming that the end user and installation of the product is unknown and that the manufacturer may or may not apply for Manufacturing Assessment.

In particular we draw your attention to the pages attached to the PDA and the restrictions that may affect its use.

- Continued to next page -

PACIFIC DIVISION

8th Floor, Kyobo Life Insurance Bldg., 7, Chungjang-daero, Jung-gu, Busan, 600-737, Republic of Korea
TEL: +82-51-469-8371, FAX: +82-51-462-4830, WEBSITE: www.eagle.org

1/4



Type Approval Certificate Extension

This is to certify that Certificate No. 11/10089(E1) for the undernoted products is extended and renumbered as shown.

This certificate is issued to:

PRODUCER	LS Cables & System Vietnam Co., Ltd.		
PLACE OF PRODUCTION	Nhon Trach 2-Loc Khang Industrial Zone, Hiep Phuoc Village, Nhon Trach District, Dong Nai Province, Vietnam		
DESCRIPTION	4 Pair LAN Cables		
TYPE	SF/UTP	Cat. 5e	HDPE, Aluminum tape coated with plastic film, Tinned copper wire braid and LSZH
	SF/UTP	Cat. 6	HDPE, X-filter (HDPE), Aluminum tape coated with plastic film, Tinned copper wire braid and LSZH
Abbreviation: HDPE : High density polyethylene insulation LSZH : Low smoke zero halogen compound jacket			
APPLICATION	Data transmission and telecommunication cabling for fixed installation in marine and offshore applications.		
DESIGN CODE STANDARD	ISO/IEC11801, IEC61156-5, IEC60332-1, IEC60754-1, IEC61034-2		
RATINGS	Impedance : 100Ω Conductor cross sectional area : AWG23 – AWG24		

Certificate No. 11/10089(E2)
Issue Date 22 September 2016
Expiry Date 21 September 2021
Sheet 1 of 2



S. Fujii
Yokohama Technical Support Office
Lloyd's Register Group Limited

Lloyd's Register Group Limited
QTA 10F, 2-3-1, Minatomirai, Nishi-Ku, Yokohama, JAPAN

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OTHER CONDITIONS

Cables do not comply with IEC60092-350, and then fixed wiring applications of the cables are to be subject to approval for each ship installation.

"This Certificate is not valid for equipment, the design, ratings or operating parameters of which have been varied from the specimen tested. The manufacturer should notify Lloyd's Register Group Limited of any modification or changes to the equipment in order to obtain a valid certificate."

The attached Design Appraisal Document No. 11/10089(E2) and its supplementary Type Approval Terms and Conditions form part of this Certificate.

All other details remain as the previous Certificate No. 11/10089(E1) to which this extension should be attached.

Certificate No. 11/10089(E2)

Issue Date 22 September 2016

Expiry Date 21 September 2021

Sheet 2 of 2

Lloyd's Register Group Limited
QTA 10F, 2-3-1, Minatomirai, Nishi-Ku, Yokohama, JAPAN

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S. Fujii
Yokohama Technical Support Office
Lloyd's Register Group Limited

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Type Approval Certificate

This is to certify that the undernoted product(s) has/have been tested with satisfactory results in accordance with the relevant requirements of the Lloyd's Register Type Approval System.

This certificate is issued to:

PRODUCER LS Cable & System Vietnam Co., Ltd.

PLACE OF PRODUCTION Nhon Trach 2-Loc Khang Industries Zone, Nhon Trach District, Dong Nai Province., Vietnam

DESCRIPTION Flame Retardant Halogen free, telecommunication cable

TYPE S/FTP PiMF LAN Cable Solid cooper wire core, polyethylene insulated, pair twisted cabling, aluminium tape wrapped, plastic film screened, overall tinned copper braid screened, LSZH sheath, flame retardant, halogen free LAN cable

APPLICATION Fixed wiring on ships and offshore installations

STANDARD Manufacture's Specifications:-
LSSS-LN0117-02(Category 6), LSSS-LN0118-02 (Category 5E)
IEC 61156-5, IEC 60332-1-1/2, IEC 61034-1/2, IEC 60754-1/2

RATINGS

Type	No. of Pairs (p)	Conductor Cross Section	Remark
Category 5E	4p	23 (AWG)	PiMF 100MHz
Category 6	4p	23 (AWG)	PiMF 250MHz

Abbreviation

PiMF: Pairs in Metal Foil

LSZH: Low Smoke Zero Halogen

Certificate No. 15/40094

Issue Date 17 August 2015

Expiry Date 17 August 2020

Sheet 1 of 2

Lloyd's Register Asia
119, Daegyo-ro, Jung-gu, Busan



G.H. LEE
Busan Technical Support Office
Lloyd's Register Asia

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LIMITATIONS OR OTHER CONDITIONS

1. If required by local regulations and indicated in the relevant detail specification, the test shall be performed in accordance with the IEC 61156-1.
2. Type Approval does not eliminate the need for normal inspection and survey procedures required by the Rules and regulation.
3. If the specified standards are amended during the validity of this certificate, the product is to be re-approved prior to it being supplied to vessels to which the amended standards apply.
4. (In case of applications other than Marine) the application of the Type Approval Certificate will be limited to the relevant scope of the Specified Standards and acceptance thereof by the local authorities.

"This Certificate is not valid for equipment, the design, ratings or operating parameters of which have been varied from the specimen tested. The manufacturer should notify Lloyd's Register Asia of any modification or changes to the equipment in order to obtain a valid certificate."

The Design Appraisal Document No.15/40094 and its supplementary Type Approval Terms and Conditions form part of this Certificate.

Certificate No. 15/40094
Issue Date 17 August 2015
Expiry Date 17 August 2020
Sheet 2 of 2

Lloyd's Register Asia
119, Daegyo-ro, Jung-gu, Busan

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G.H. LEE
Busan Technical Support Office
Lloyd's Register Asia

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DNV GL

Certificate No:
TAE000014X

TYPE APPROVAL CERTIFICATE

This is to certify:

That the Fiber optical cable

with type designation(s)
QFCI, QFCU, AICI

Issued to

LS Cable & System Co., Ltd. (Indong Plant)
Gumi-si Gyeongbuk, Republic of Korea

is found to comply with

Det Norske Veritas' Rules for Classification of Ships, High Speed & Light Craft and Det Norske Veritas' Offshore Standards
IEC 60332-3-24 (2009-02)
IEC 60754-1/2 (2011-11)
IEC 60684-2 Ed. 3 (2011-08)
IEC 61034-2 Ed. 3.1 (2013-06)

Application :

Product(s) approved by this certificate is/are accepted for installation on all vessels classed by DNV GL.

Fiber optic cable. Flame retardant Cat. C. Fire resistant (QFCI & QFCU). Halogen free. Low smoke. Mud resistant (QFCU)

This Certificate is valid until **2021-07-04**.

Issued at **Busan** on **2016-07-05**

DNV GL local station: **Ulsan**

Approval Engineer: **Eun Sook Kim**



Baeg Soon Choi
Head of Section

This Certificate is subject to terms and conditions overleaf. Any significant change in design or construction may render this Certificate invalid. The validity date relates to the Type Approval Certificate and not to the approval of equipment/systems installed.

Form code: TA 1411a Revision: 2015-05
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www.dnvgl.com

Page 1 of 4

KOREAN REGISTER

KR

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TYPE APPROVAL CERTIFICATE

Certificate No.

: POH25526-EL001

Initial Approval : 7th August, 2017

Product

: Electric Cable

Manufacturer

: LS Cable & System Ltd., Gumi Plant

228, Suchul-daero, Gumi-si, Gyeongbuk-do, 39369, Republic of Korea

Product Description : Telecommunication Cable

[Type : 4Pair SF/UTP Cable with LSZH Jacket(Enhanced Category 5)]

" Place of production "

Tai Young Electronics Co., Ltd.

- 29, Gongdan 1-daero 259beon-gil

Siheung-si, Gyeonggi-do, 15079, Republic of Korea

" See Appendix 1 "

Approval Condition : " See Appendix 1 "

THIS IS TO CERTIFY

that the above-mentioned product has been approved in accordance with the relevant requirement of this Society's Rules and / or of the recognized standards as follows and entered in the "List of Approved Manufacturers and Type Approved Equipment".

Pt. 6, Ch. 1, Sec. 4 of the Rules for Classification, Steel Ships and IEC 11801, 61156-5 IEC 60754-2, 61034-2, 60332-1-2.

This Certificate is valid until 6th August, 2022

Issued at Busan, Korea on 7th August, 2017

KOREAN REGISTER OF SHIPPING

General Manager of

Marine & Ocean Equipment Team

Note : 1. This certificate will be valid subject to complying with the approval conditions described on the certificate and/or on the Rules of this Society.

2. This certificate will be invalid from the expiry date aforementioned unless the extension or renewal has been granted to the applicant or the manufacturer.

3. Any significant modifications or changes in design or construction to the above product without approval from this Society will render this certificate invalid.

4. Should the specified rules, regulations or standards be amended during the validity of this certificate, the product is to be re-approved by this Society in accordance with the requirements as amended.

● 기타

RCDD	82
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RCDD (Registered Communication Distribution Designer)

RCDD

미국의 최대 통신하드웨어 설치 및 관리 전문위원회인 BICSI(Building Consulting Service International)에서 주관하는 자격 인증 제도로 UTP, 광 및 무선통신을 포함하는 포괄적인 통합배선시스템의 설계, 운영, 법규 규제 등에 관한 전문기술자 자격을 의미함.



Simple Warranty

Warranty Program



프로그램 개요

지속적인 사후 관리를 통해 제품 신뢰성을 제공하고, 고객 만족을 향상시키기 위한 프로그램
LS Simple™ Solution의 기술력에 대한 자신감을 바탕으로 품질 보증 제공

운영

LS Authorized SI업체로 등록된 업체가 시공한 경우에 발행
LS Installation 교육 프로그램을 수료한 인원이 시공에 참여하여,
표준에 근거한 설계 및 설치 기법 적용하고, LS 전선이 설치 상태를 확인한 후에 품질 보증서 발급
천재 지변 및 사용자 부주의의 경우, 품질 보증에서 제외

사후 관리

불만 사항 접수 시, 신속한 원인 파악 및 체계적인 대응으로 긴급 복구 체계 제공
제품 불량 등의 발생 시 무상 교체 진행

LS Certified Installer Program



프로그램 개요

LS전선의 제품을 신뢰성 있게 설치할 수 있도록, 교육을 이수한 공사업체에게 제공하는 프로그램
Warranty 발급 전에, LS전선을 통하여 이론 및 설치 실습 교육을 이수하도록 하여 완성도 있는 공사를 수행할 수 있도록 지원

접수 방법

PJT 수행시기에 LS전선 본사 통합배선영업팀을 통하여 LSI 신청 문의

주요 건설사



납 품 실 적 증 명 원

제조사	주 소	경기도 안양시 동안구 엘에스로 127, LS타워 12~17층		
	상 호	LS전선(주)	전 화 번 호	02-2189-9114
	사업자등록번호	138-81-47634	대 표	명노현
용 도		실적 증명	부 서 명	통합배선영업팀
아래와 같이 납품한 실적이 있음을 증명합니다.				
● Solution Name: LS전선 통합배선 Total-Solution ● 적용 현장: 공공기관 및 공공시설, 병원/호텔, 학교 ● Solution 제품상세: 1. 통신케이블: UTP 케이블, STP 케이블, 광케이블, 동축케이블 2. 전력케이블: HFIX, HFCO, F-CVV-SB, F-GV 3. 통합배선자재: 110 Block, Patch Panel, Faceplate, FDF, UTP/STP 패치코드, Modular Jack F/O Jumper Cord				
Category	납품 현장	Category	납품 현장	
공공기관 및 공공시설	국회의원회관	병원/호텔 & 학교	영종 메디컬센터	
	마포경찰공제회		창원 경상대 병원	
	대구혁신도시		대전 서부 병원	
	중앙신체검사소		평택 미군부대 초고교	
	대전 핵융합 연구원		부산 파라다이스 호텔	
	서울시 청사		서울여대 50주년 기념관	
	충남도청 신도시		부산 동서대학교	
	경북 첨단의료 복합단지		울지 대학교	
	대전 서부 병원		한국종합예술학교	
	인천도시계획관		인천 성모 병원	
	황금박쥐(Army)		단국대 천안 Campus	
	서조산업(Army)		연세대 송도 Campus	
	인천 연수구청		울산 과학기술 대학교	
	상암동 사이버안전센터		서강대학교	
	방배행정문화센터		광화문 포시즌 호텔	
	경북 첨단의료 복합단지		부산대 병원 외상센터	
	인천국제공항 3단계			
위의 사실을 증명합니다.				
LS전선(주) 대표이사 명노현				



납 품 실 적 증 명 원

제조사	주 소	경기도 안양시 동안구 엘에스로 127, LS타워 12~17층		
	상 호	LS전선(주)	전 화 번 호	02-2189-9114
	사업자등록번호	138-81-47634	대 표	명노현
용 도		실적 증명	부 서 명	통합배선영업팀
아래와 같이 납품한 실적이 있음을 증명합니다.				
● Solution Name: LS전선 통합배선 Total-Solution ● 적용 현장: Data Center, 기업형 빌딩 ● Solution 제품상세: 1. 통신케이블: UTP 케이블, STP 케이블, 광케이블, 동축케이블 2. 전력케이블: HFIX, HFCO, F-CVV-SB, F-GV 3. 통합배선자재: 110 Block, Patch Panel, Faceplate, FDF, UTP/STP 패치코드, Modular Jack F/O Jumper Cord				
Category	납품 현장		납품 현장	
Data Center	SK Telecom 수유국사	IBS	네이버 지식정보 캠퍼스	
	SK Telecom 성수 국사		김천한전기술 사옥	
	송도 IBM 전산센터		전경련회관	
	부산 미음 전산센터		LG 대전기술연구원	
	부평 현대해상 전산센터		송도 H2호텔	
	국방부 메가데이터 센터		판교 네오위즈 사옥	
	우리은행 종합지원센터		청진동 GS종로타워	
	충주 포스코 DataCenter		한국은행 인재개발원	
IBS	대검찰청 전산센터		평촌 GS 스퀘어	
	상암 IT Complex		LG U+ 용산사옥	
	상암동 LG U+ 사옥	NHN지식정보 캠퍼스		
	이천 SK 하이닉스	KT청진사옥		
	인천 V-ENS 산업단지	SK텔레콤 부암사옥		
	IFC Seoul	롯데시티 호텔제주		
	부산 메리츠화재 사옥	LG 사이언스파크		
	송도 I-Tower	가산 LG전자 R&D 센터		
	진주 LH 사옥	현대차 의왕 연구소		
위의 사실을 증명합니다.				
LS전선(주) 대표이사 명노현				

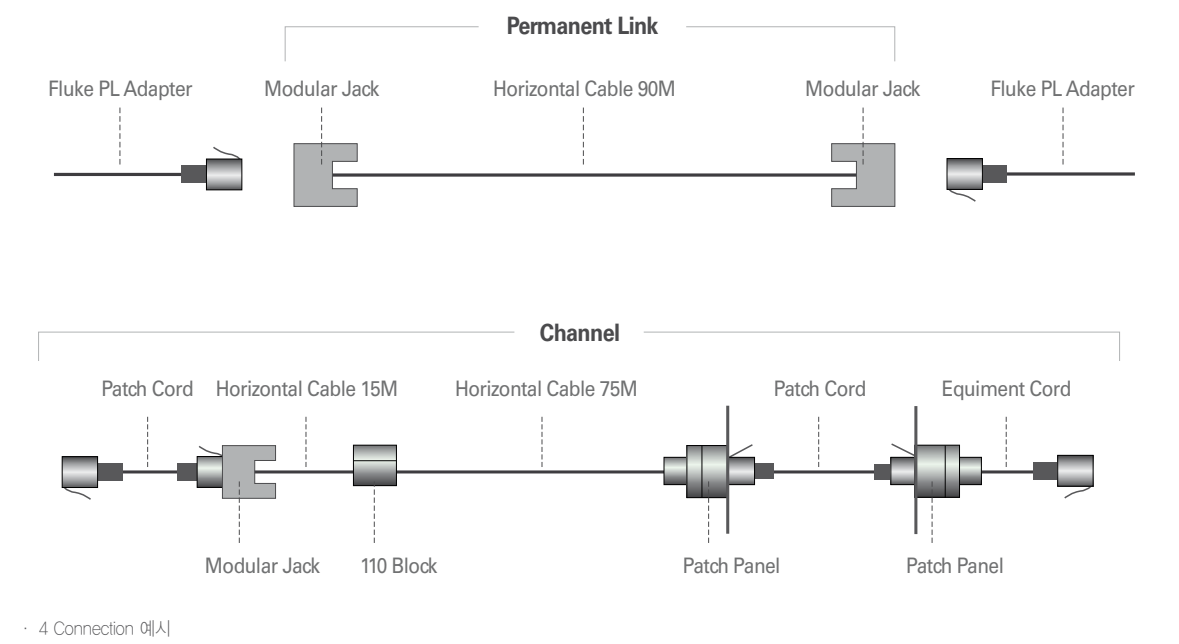
난연등급별 사용용도

등급이 높을수록 화재시 케이블에 의한 인명 피해 최소화				
구분	CMX	CM	CMR	CMP
난연별 사용용도	일반 주거용도 사용	층내 수평 배선용도 사용	층간 수직 배선용도 사용	층내 고난연 배선용도 사용
관련규격 및 규제사항	• Non-Plenum Cable • UL 1581 (VW-1 Test) • IEC 60332-1	• Non-Plenum Cable • UL 1685 (VTFT Test) • IEC 60332-3	• Non-Plenum Cable • UL 1666 (Riser Test)	• Plenum Cable • NFPA 262 • 연기밀도 규제
Sheath	PVC LSZH	PVC LSZH	PVC LSZH	PVC

· 유럽 – 건축자재에 대한 CPR 인증으로 연소 길이, 발생열량, 무독성, 연기밀도, 불꽃 발생 여부를 규제함 (LS는 LSZH 자재로 Cca, B2ca 등급 Level까지 공급 가능)

· 미국 –CMR/CMP 급 PVC (CMP의 경우 화재 발생시 Sheath 및 절연체의 연소가 없음)

Permanent Link & Channel



주의사항

- UTP Cable을 설치할 때에는 주어진 신호전송을 위하여 Cable, Patch Cord, Patch Panel, Outlet 자재등을 동일한 Category로 사용하여야 한다.
- 수평 Cable 4Pair 포설시 도체 또는 꼬임 길이의 늘어남을 방지하기 위하여 최대 인장력은 110N를 초과하지 않아야 한다.
- 수평 4P Cable 굴곡시 최소 곡률 반경은 25mm(1inch : 통상 Cable 의 4배) 이상 유지해야 한다.(4 Pair 기준)
- 배선자재 연결시 최대 외피 제거 권장 길이는 최대 40mm를 유지하여 가능한 Cable 구조를 유지해야 한다.
- 습기는 전기적 특성에 있어서 치명적인 장애요인으로 비나 습기에 노출되지 않는 곳을 선택하여 보관하여야 한다.
- UTP Cable 설치시 TIA/EIA 기준에 따라 적절한 이격 거리를 유지해야 한다.

간선 및 실내케이블에 대한 이격거리

구분	A-wire	최소이격거리
전력선	특별고압가공전선 (단, 특별고압가공전선이 케이블이고 통신선이 절연전선인 경우)	1,2m 30cm
	고압, 저압가공전선 (단, 고압, 저압가공전선이 케이블이고 통신선이 절연전선인 경우)	60cm 30cm
라디오, TV	보호, 접지된 독립배관의 선 안테나, 접지선	— 10cm
	CATV 접지	접지 차폐된 동축케이블
네온사인	변압기로부터 인출된 선	15cm
형광등	피뢰침 및 피뢰선	1,8m

카테고리 별 전송특성 비교표

구분	전송능력	대역폭	케이블 Type	Pair 수
Category5	100Mbps	100 MHz	U/UTP	4, 25, 50, 100P
Category5E	400Mbps	100MHz	U/UTP, F/UTP, SF/UTP, S/FTP,F/FTP	4, 25P,1Q
Category6	1Gbps	250MHz	U/UTP, F/UTP, SF/UTP, S/FTP,F/FTP	4P
Category6A	10Gbps	500MHz	U/UTP, F/UTP, S/FTP,F/FTP	4P
Category7	10Gbps이상	600MHz	S/FTP,F/FTP	4P

GLOBAL NETWORK

More than 60 Factories,
Sales and Production Sites
in 20 Countries.

- Factory
- Sales office
- Branch office



KOREA

Gumi Plant
EHV / MV / LV cable
UTP, Coaxial cable
SCR, Magnet wire
Overhead cable, Bus duct


Indong Plant
Optical fiber
Optical cable


Donghae Plant
Submarine cable
Industrial specialty cable

CHINA

LSHQ(Yichang)
EHV / MV / LV cable
Industrial specialty cable


LSCW(Wuxi)
Industrial devices cable
Automotive cable
Harness & module
Aluminum, Bus duct

VIETNAM

LS-VINA(Haiphong)
EHV / MV / LV cable
SCR, ACSR
Overhead cable


LSCV(HO Chi Minh)
MV / LV cable
UTP, Optical cable
Overhead cable

INDIA

LSCI(Bawal)
EHV / MV / LV cable
Coaxial cable
Overhead cable

USA

LSCUS(Tarboro)
MV / LV cable
Control, Instrument cable

POLAND

LS EV Poland./LSCP (Dzierzoniow)
Automotive battery components
Optical cable



Simple™

www.lssimple.co.kr
Simple™ Structured Cabling System

04386, 서울특별시 용산구 한강대로 92 LS용산타워 19층
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